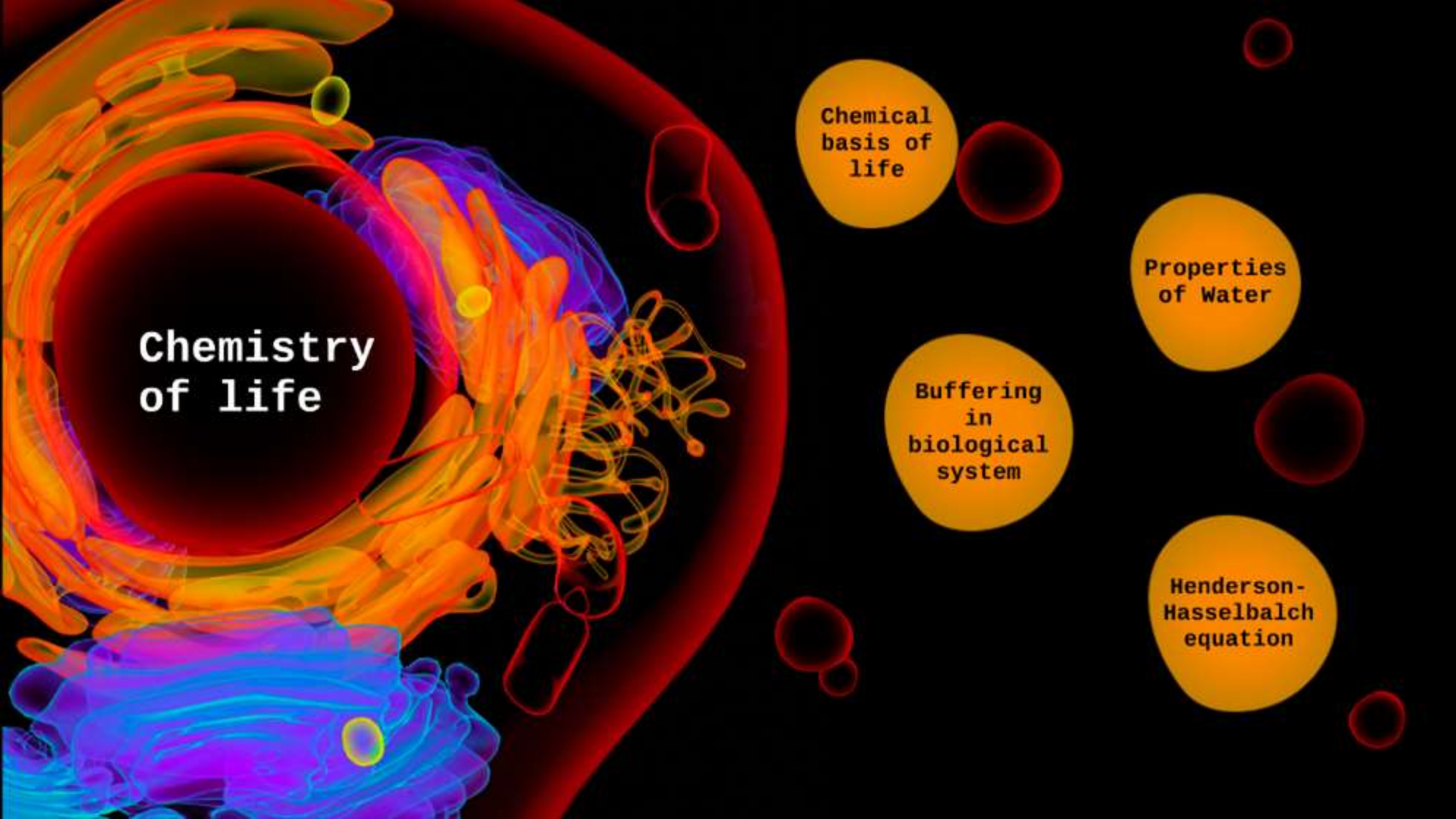
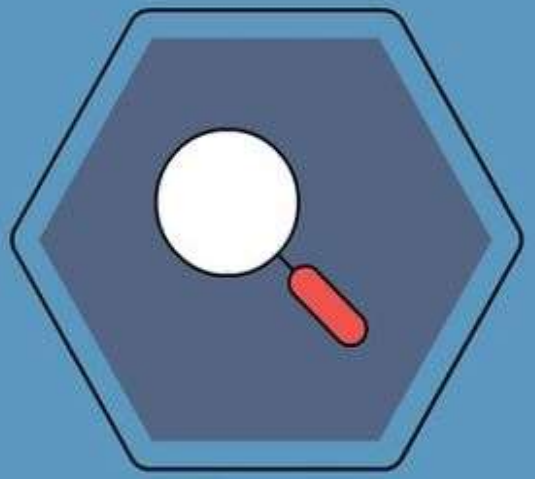

Chemistry of life
Chemical basis of life
Properties of Water
Buffering in biological system
Henderson-Hasselbalch equation

4 elements make up over 98 % of all biological matter:

C 18%

CARBON

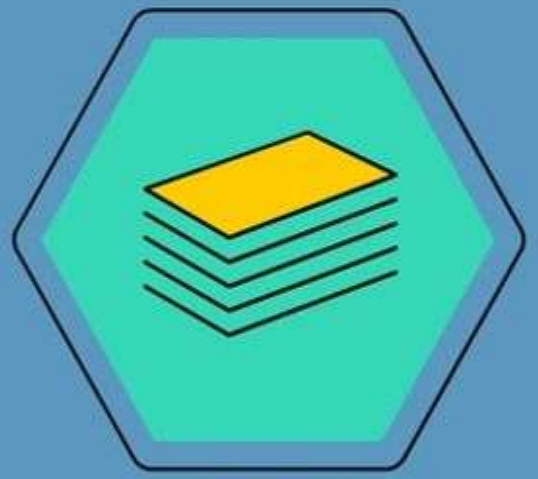


O 65%

OXYGEN

H 10%

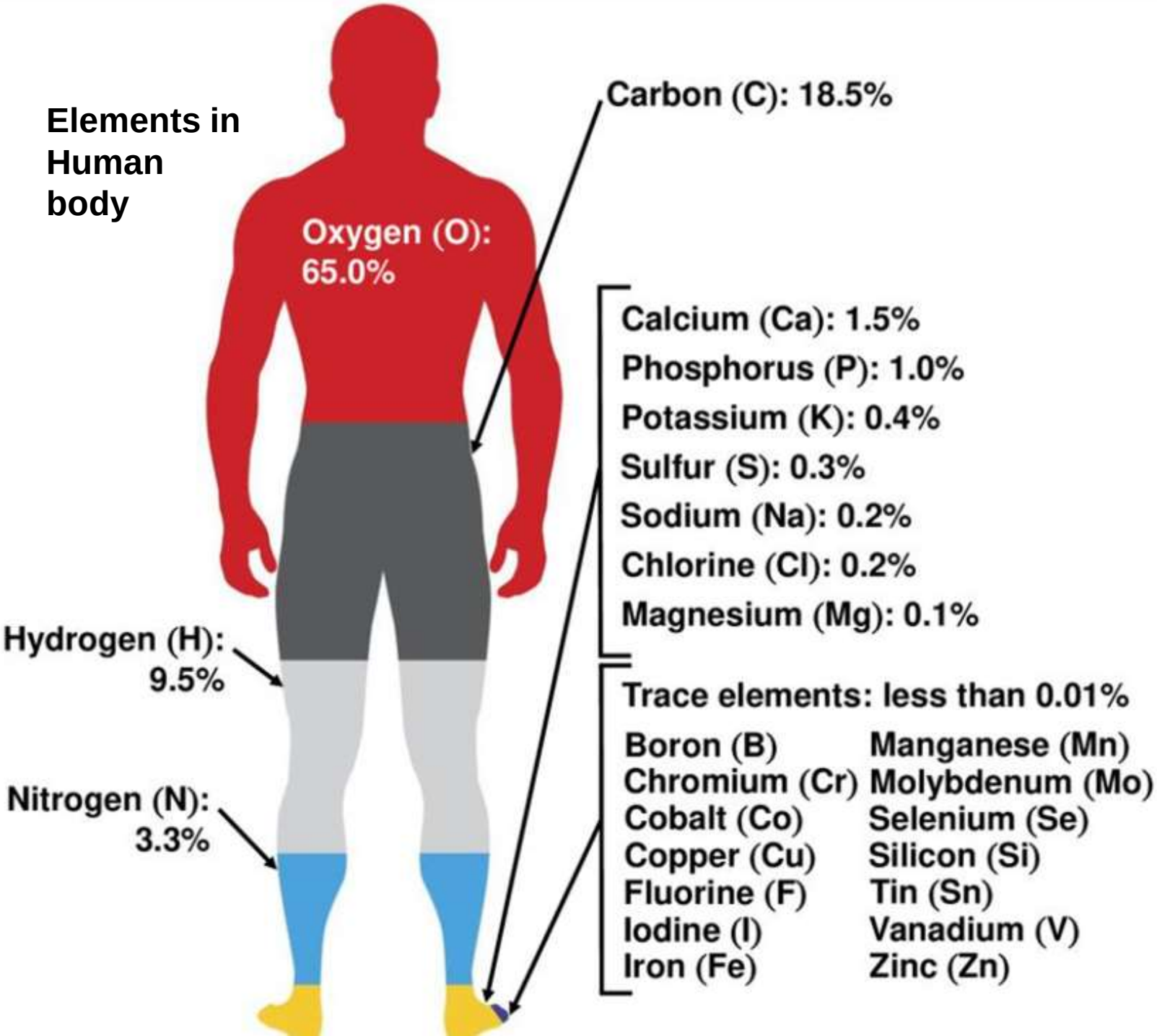
HYDROGEN



N 3%

NITROGEN

Elements in Human body



The percentage of cell molecules by mass is:

- **Water**: 70–90%
- **Proteins**: 10–15%
- **Nucleic acids**: 5–7%
- **Carbohydrates**: 3%
- **Lipids**: 2%
- **Ions**: 1%

The macromolecular composition of a cell by dry weight is:

- **Proteins**: About 55%
- **RNA**: About 20%
- **Lipids**: About 10%
- **DNA**: About 3%
- **Polysaccharides**: The rest
- **Metabolites**: The rest
- **Ions**: The rest

The major and minor elements of life refer to the essential chemical elements that are necessary for life processes.

Major Elements

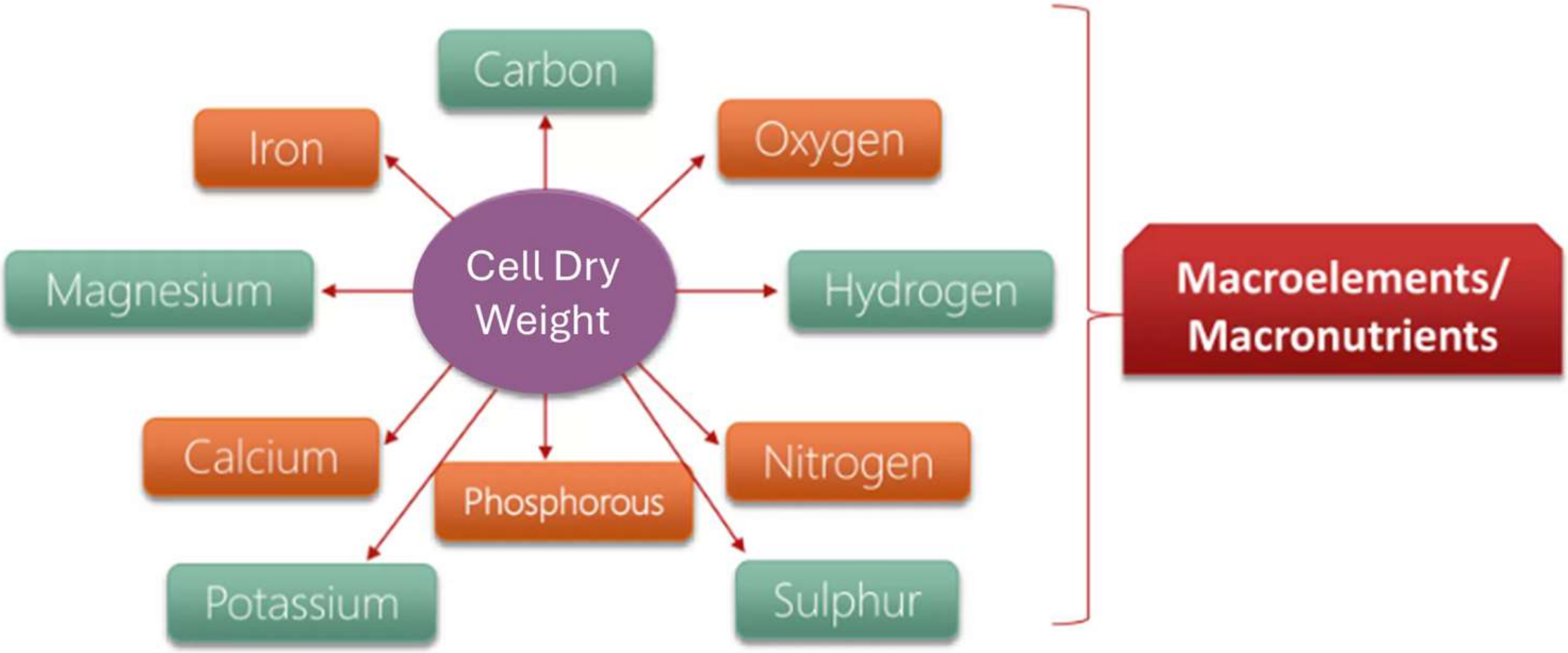
These elements make up the majority of living organisms and are crucial for life

1. **Carbon (C)**
2. **Hydrogen (H)**
3. **Oxygen (O)**
4. **Nitrogen (N)**
5. **Phosphorus (P)**
6. **Sulfur (S)**

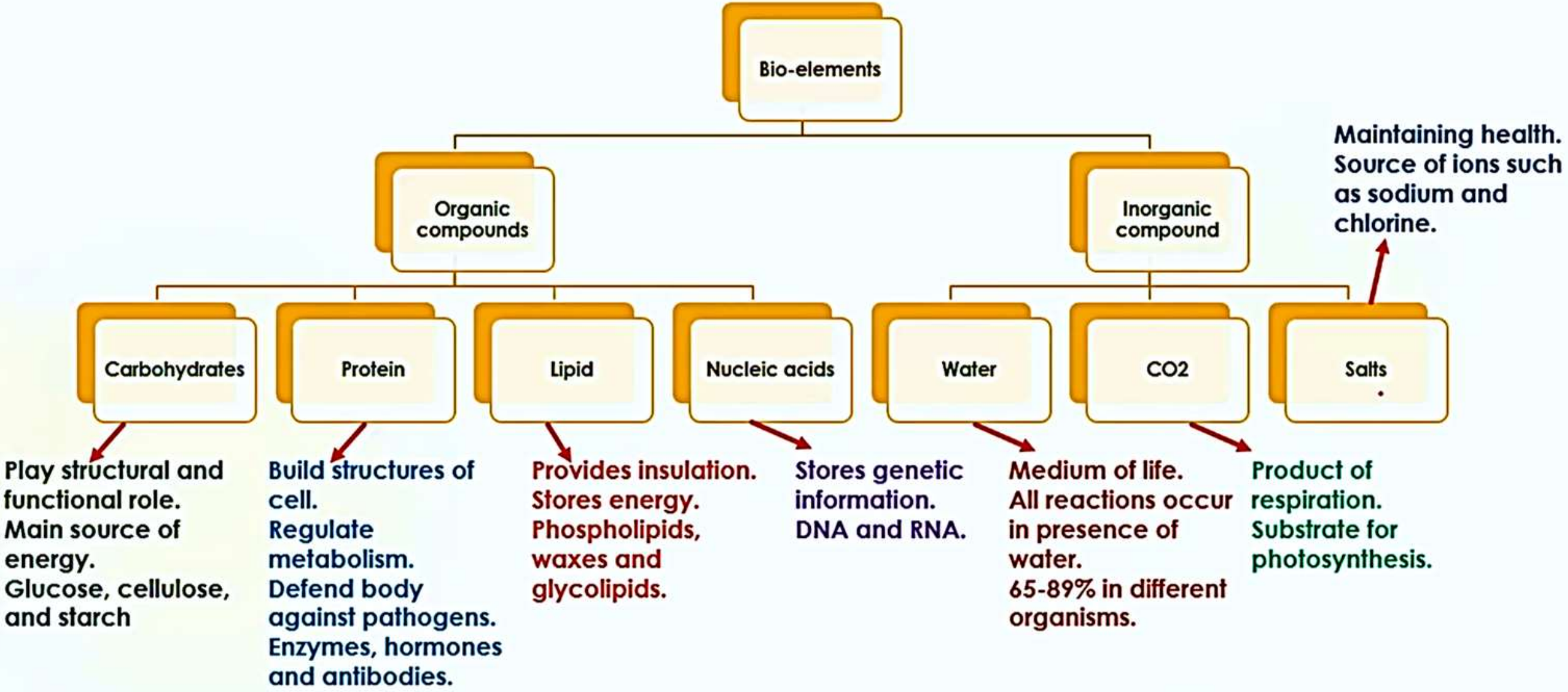
Minor Elements

These elements are required in smaller amounts but are still essential for various biological functions

1. **Calcium (Ca)**
2. **Potassium (K)**
3. **Sodium (Na)**
4. **Magnesium (Mg)**
5. **Iron (Fe)**
6. **Zinc (Zn)**
7. **Iodine (I)**


Carbon
Iron
Oxygen
Cell Dry Weight
Magnesium
Hydrogen
Calcium
Phosphorous
Nitrogen
Potassium
Sulphur
Macroelements/ Macronutrients

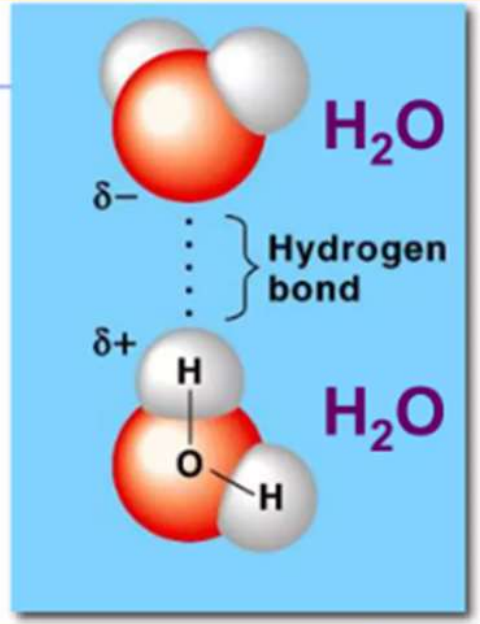
Chemical composition of cell



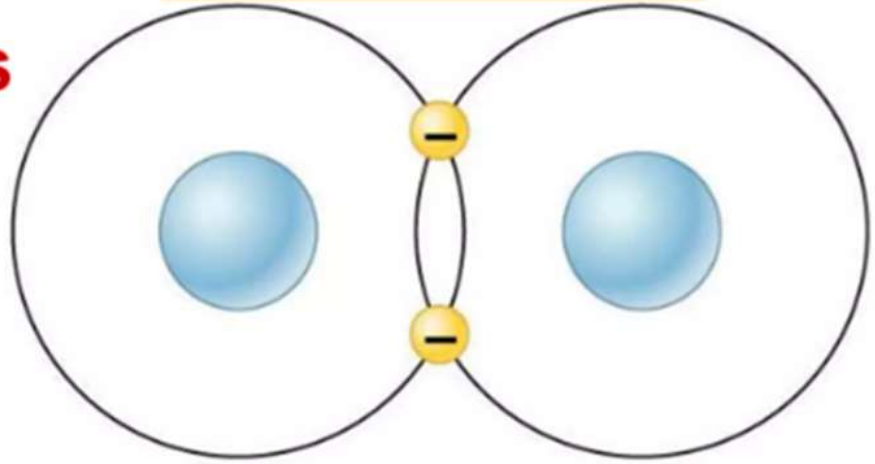
Bonds in Biology

- **Weak bonds**
  - hydrogen bonds
    - attraction between + and –
  - hydrophobic & hydrophilic interactions
    - interaction with H_2O
  - van der Waals forces
  - (ionic)
- **Strong bonds**
  - covalent bonds

Hydrogen bond



Covalent bond



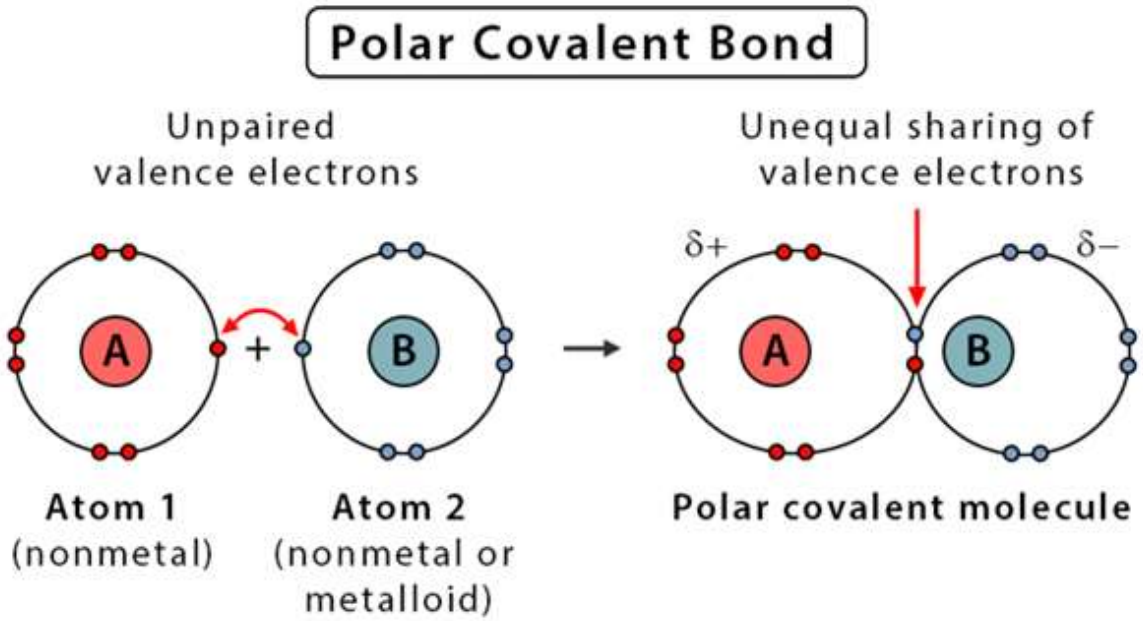
H_2 (hydrogen gas)

Bond Type	Bond formation
Ionic	Electrostatic force between cation and anion
Covalent	Electron sharing at outer shell

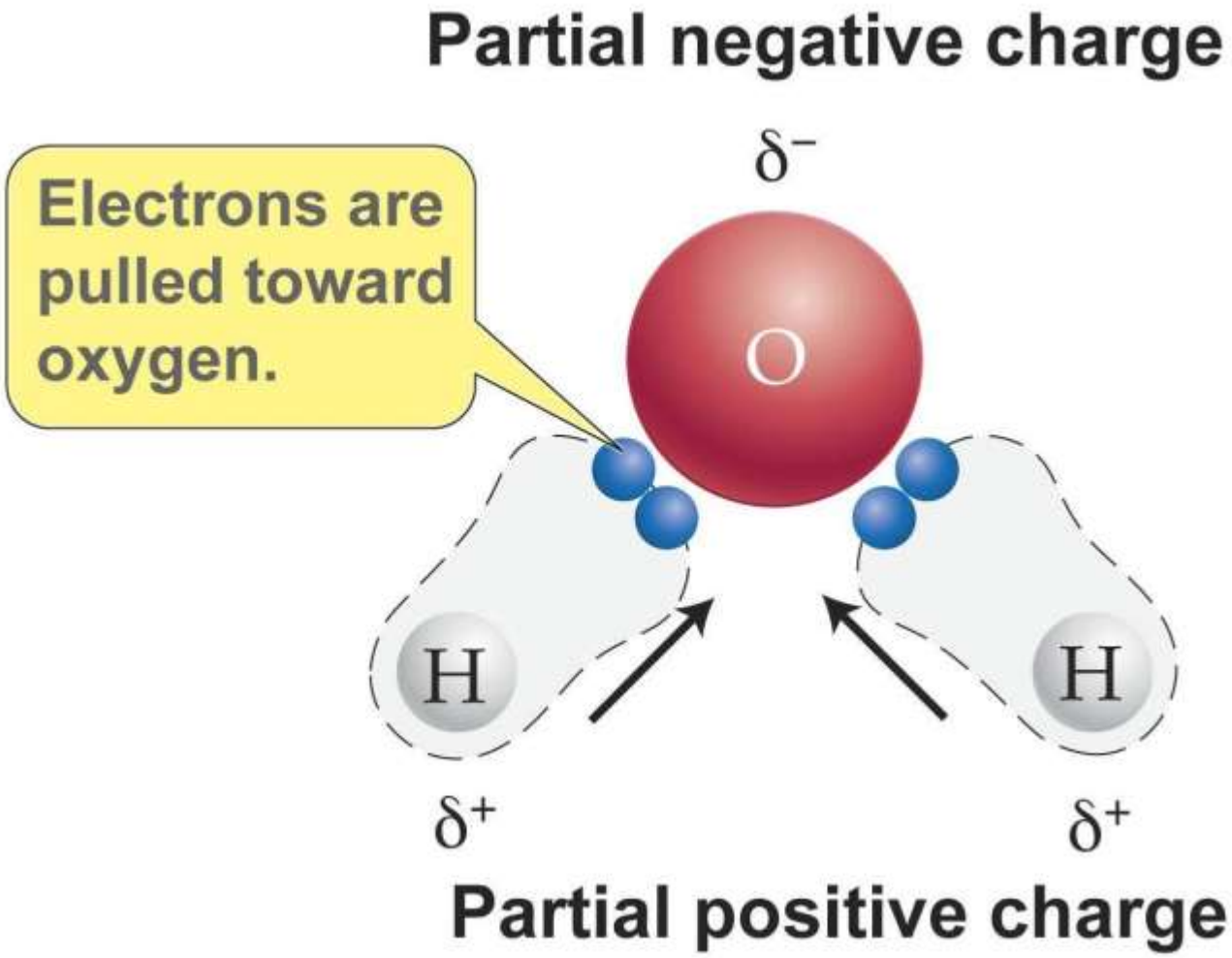
Electronegativity

Electronegativity is a measure of the ability of an atom in a molecule to draw bonding electrons to itself.

A **polar covalent bond** is a covalent bond in which the bonding electrons spend more time near one atom than the other.



Polar covalent bond in water



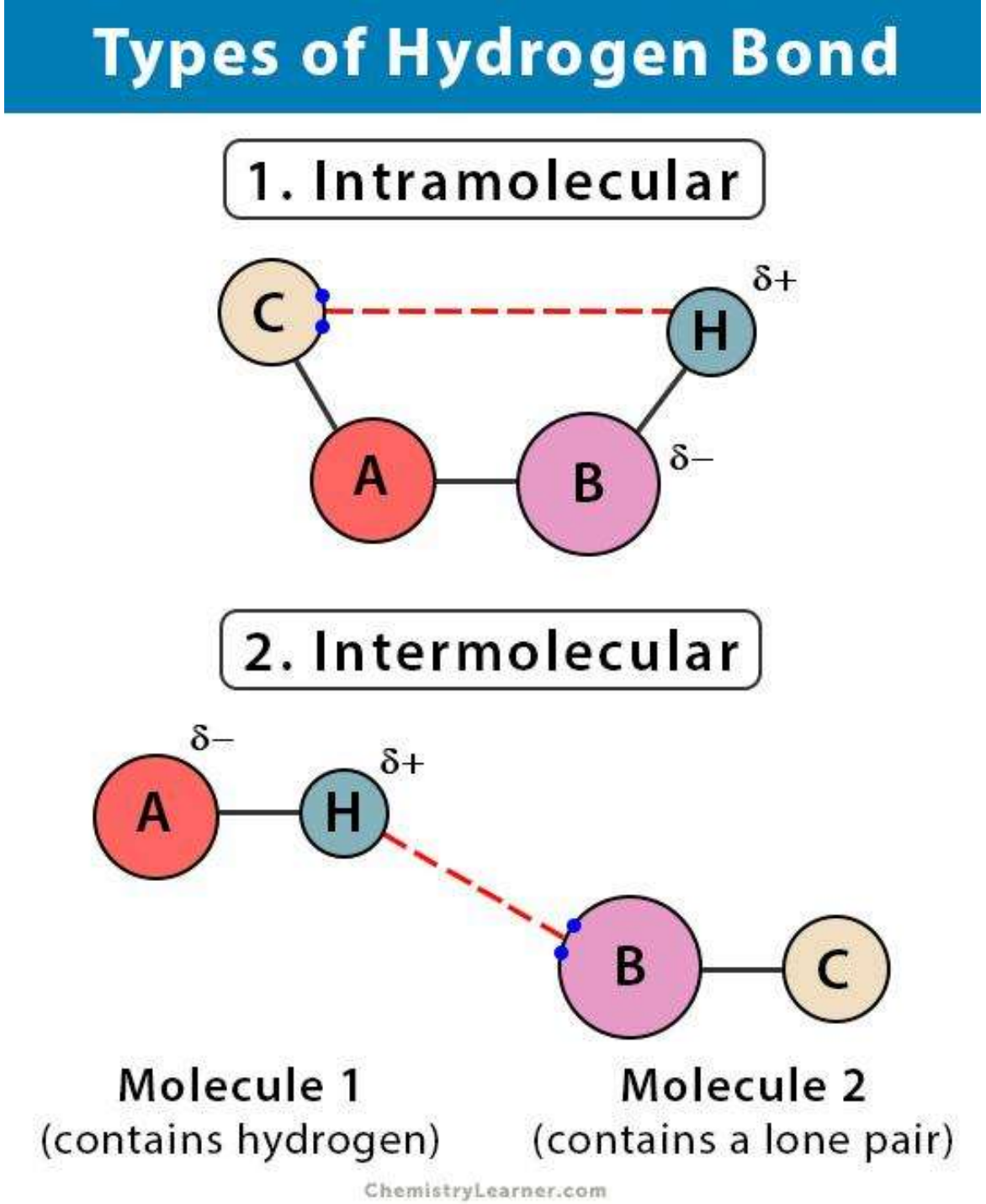
HYDROGEN BOND

Hydrogen bonding is a weak to moderate attractive force that exists between a hydrogen atom covalently bonded to a very electronegative atom, X, and a lone pair of electrons on another small, electronegative atom, Y.

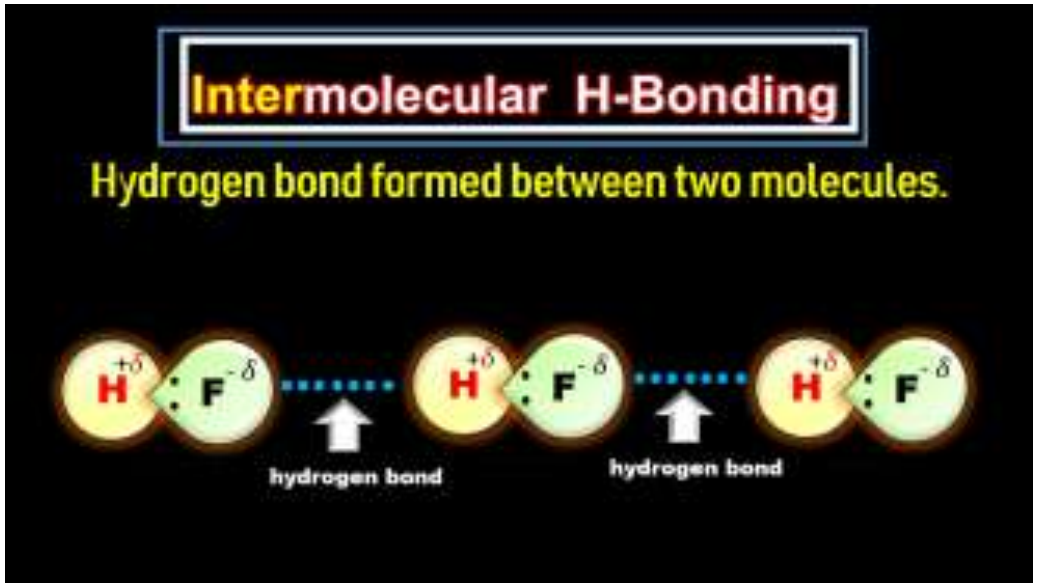
Molecules held together by **polar covalent bonds** has no **NET charge**

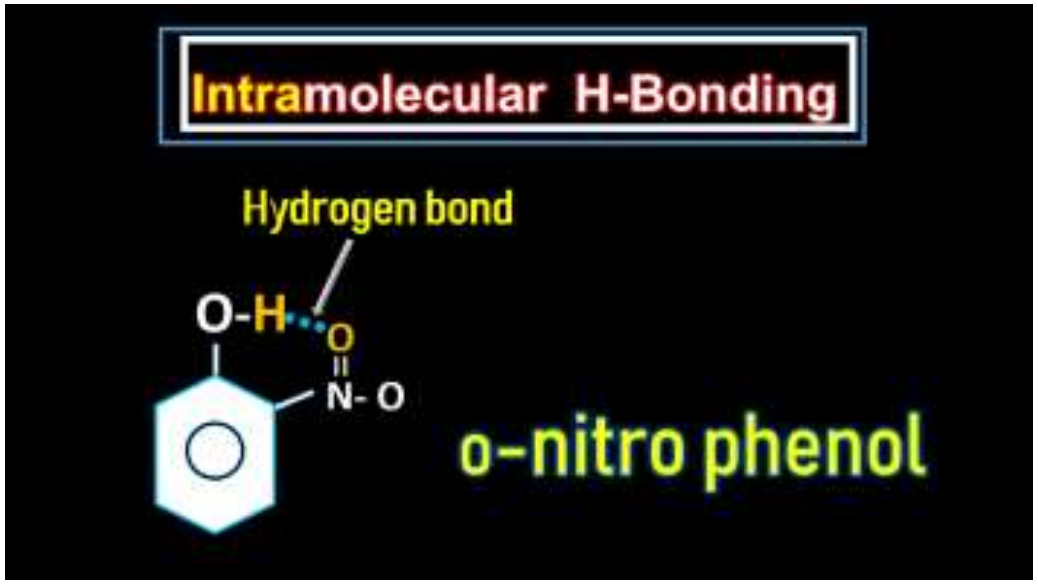
Hydrogen bond is two types

- Intermolecular Hydrogen Bond (e.g. water)
- Intramolecular Hydrogen Bond (e.g. o-nitrophenol)



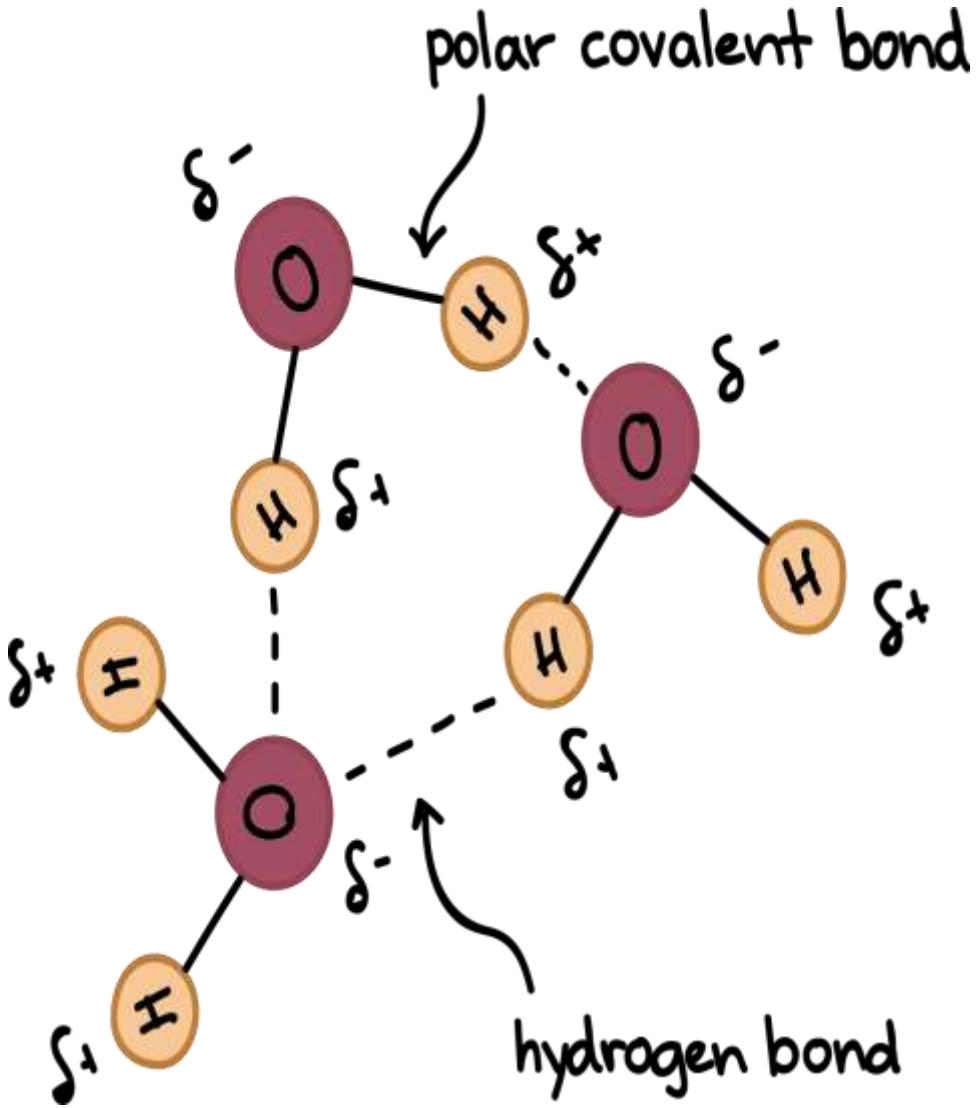
HYDROGEN BOND





Hydrogen Bond in Water

- Water is a polar covalent compound
- Water Molecule has no net charge
- But Oxygen has a strong attraction for electrons than hydrogen.
- They pull the shared electron towards them as a result they form a partial negative charge.
- And Hydrogen end has a slight positive charge
- Which results, hydrogen bonds formation between neighboring hydrogen and oxygen atoms of adjacent water molecules.



Why Ice Floats

- Ice has a crystalline structure formed with the help of hydrogen bonds, where each water molecule is hydrogen-bonded to four neighboring water molecules.

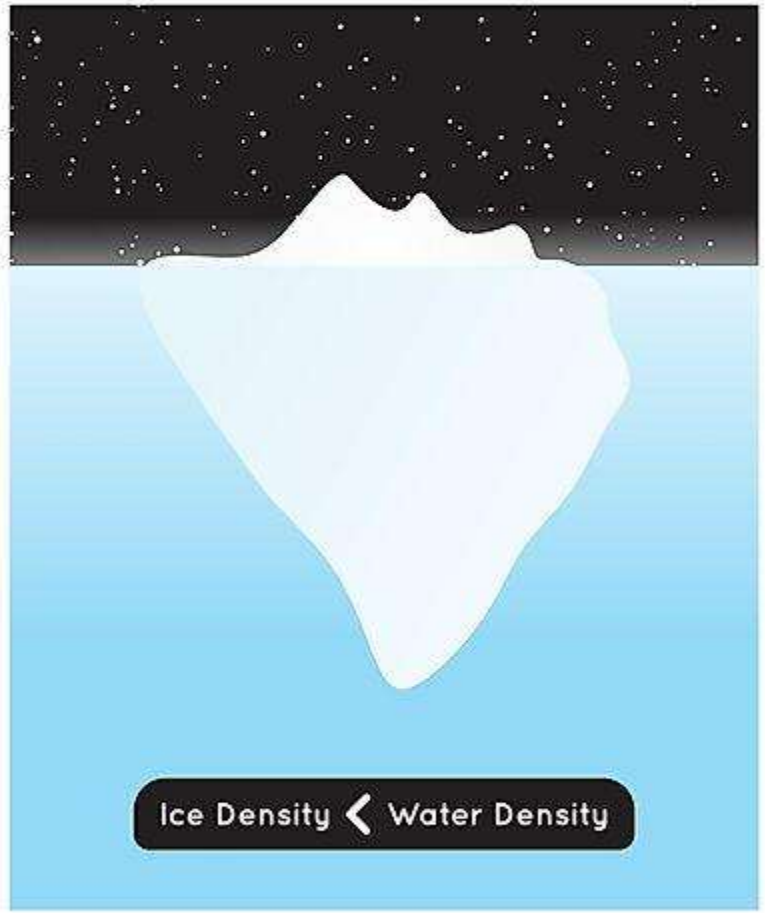
- These hydrogen bonds are stable, creating spaces between the molecules, which leads to a decrease in density.

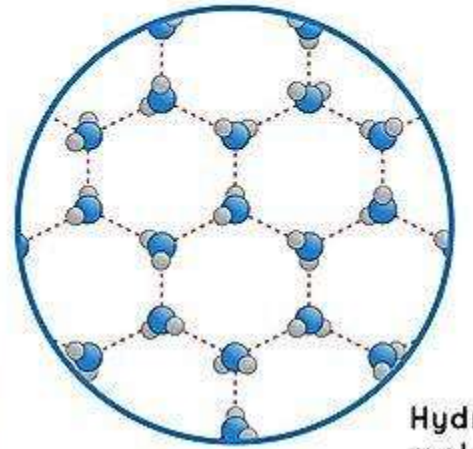
- In liquid water, hydrogen bonds are more dynamic, allowing molecules to be more closely packed.

- The tetrahedral arrangement of hydrogen bonds in ice results in an open structure with hexagonal channels, making ice less dense than water.

- When ice melts, some hydrogen bonds break, and the structure collapses, increasing the density.

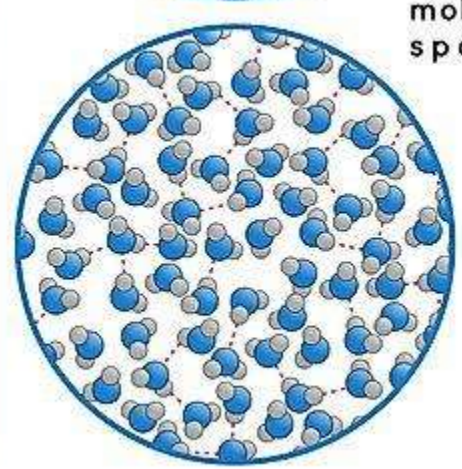
Why Does Ice Floats On Water?





Ice Molecules

Hydrogen bonds between ice molecules allow them to be spaced further apart



Water Molecules

Importance of Hydrogen Bonds

- Source of unique properties of water
- Structure and function of proteins
- Structure and function of DNA
- Structure and function of polysaccharides
- Binding of a substrates to enzymes
- Binding of hormones to receptors
- Matching of mRNA and tRNA

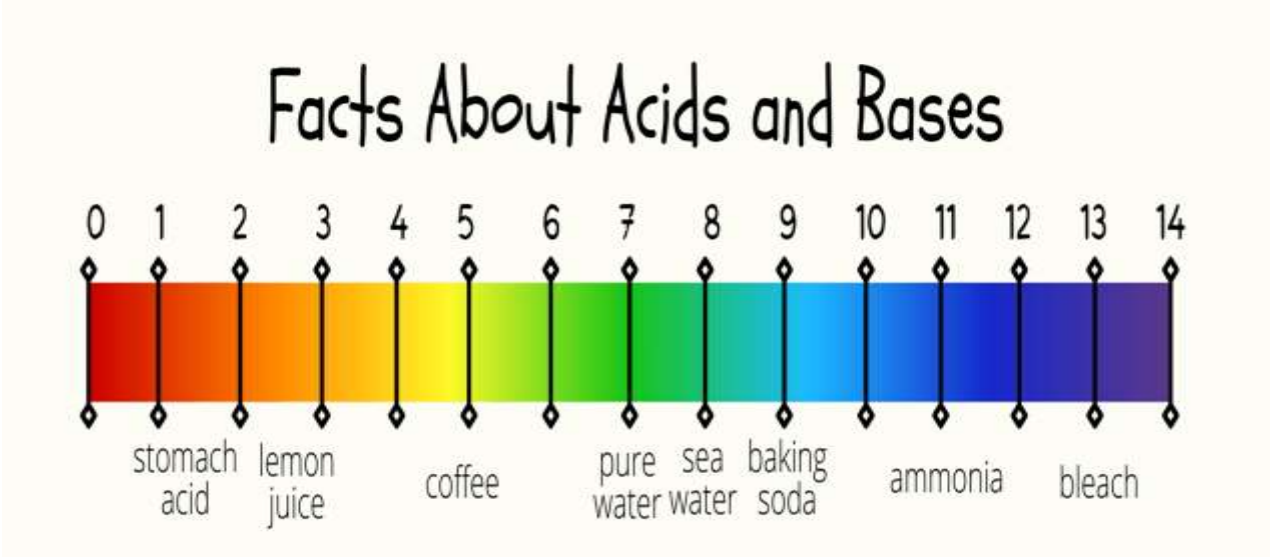
pH

- **pH** is defined as the negative logarithm of H^+ ion concentration.

  pH = -log [H+]

- A pH scale is a tool for measuring acids and bases.
- The scale ranges from 0-14

pH Range	Solution
0 - <7	Acidic
~7	Neutral
>7 - 14	Basic



Acids

Donate H^+ when dissolved in water

Example: HCl

Bases

Accept H^+ when dissolved in water

Example: NH_3

Buffer Solution

A buffer is a solution characterized by the ability to resist changes in pH when limited amounts of acid or base are added to it.

weak acid + its conjugate base

Or

weak base + its conjugate acid

Acid - H^+ = conjugate base

Base + H^+ = conjugate acid

Buffers are broadly divided into two types: acidic and alkaline buffer solutions.

Acidic buffers are solutions that have a pH below 7 and contain a weak acid and one of its salts. For example, a mixture of acetic acid and sodium acetate acts as a buffer solution with a pH of about 4.75.

Alkaline buffers, on the other hand, have a pH above 7 and contain a weak base and one of its salts.
For example, a mixture of ammonium chloride and ammonium hydroxide acts as a buffer solution with a pH of about 9.25

Buffer in Human Body

One example of a buffer solution found in nature is blood. The body's acid–base balance is normally tightly regulated, keeping the arterial blood pH between 7.38 and 7.42.

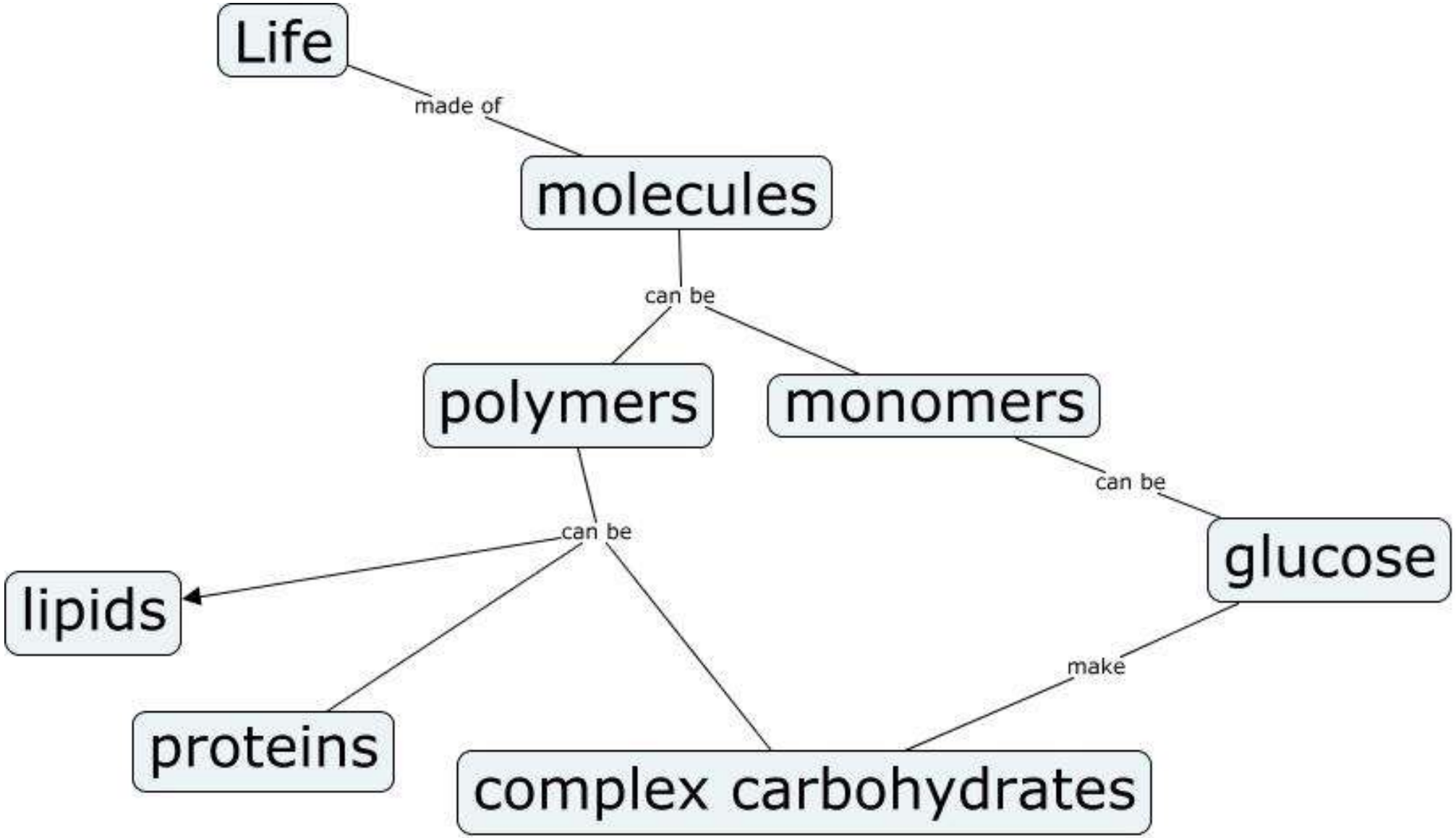
$$CO_2 + H_2O \longleftrightarrow H_2CO_3 \longleftrightarrow H^+ + HCO_3^-$$

Or, simply

$$CO_2 + H_2O \longleftrightarrow H^+ + HCO_3^-$$

pH drop → H^+ level drops → equilibrium shifts to right → H^+ level restored

pH rise → H^+ level rises → equilibrium shifts to left → H^+ level dropped to normal


Life
made of
molecules
can be
polymers
monomers
can be
glucose
can be
lipids
make
proteins
complex carbohydrates

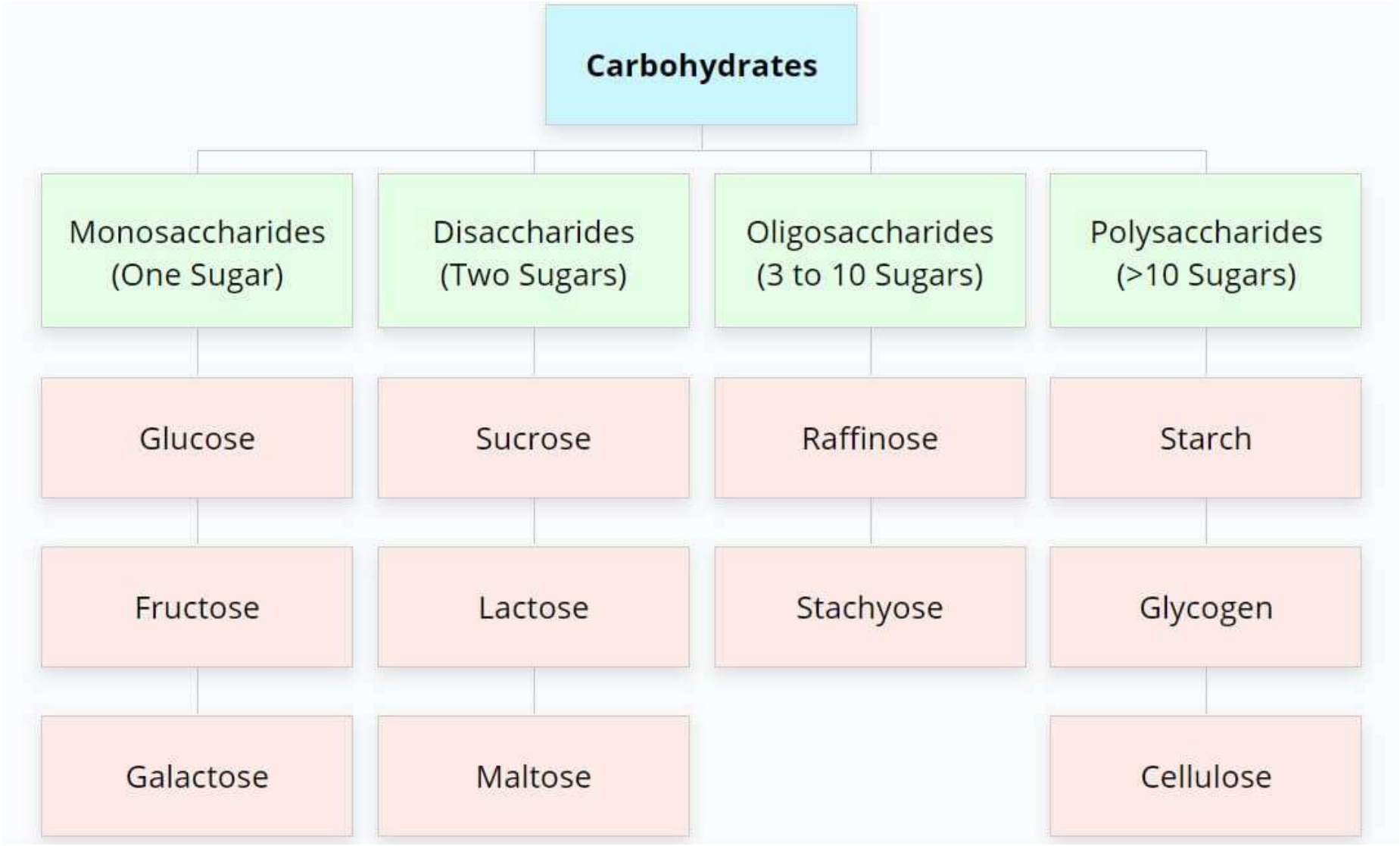
Biomolecules

- Biomolecule are molecules and ions present in organisms that are essential to one or more biological processes.
- Many of the biomolecules are polymers.
- A polymer is a compound made up of several repeating units (monomers) and produced by polymerization.
- Most of these biomolecules are organic compounds.
- Being “organic” means, in general, they contain carbon atoms covalently bound to other atoms, especially Carbon-Carbon (C-C) and Carbon-Hydrogen (C-H).
- The four major element constituents are carbon, hydrogen, oxygen, and nitrogen.

Carbohydrates

- A carbohydrate is a bio-molecule consisting of carbon (C), hydrogen (H) and oxygen (O) atoms.
- **Saccharide** is the structural (monomeric) unit of carbohydrates. Based on the number of monomeric units, carbohydrates may be monosaccharides, disaccharides, oligosaccharides, or polysaccharides.
- Carbohydrates perform numerous roles in living organisms. They can storage energy(e.g. starch and glycogen), act as structural components (e.g. cellulose in plants) and even form the backbone of the genetic molecule known as RNA.
- 1 gram of carbohydrate gives 4 Kcal (approx.)

Classification of Carbohydrates



Glycosidic Bond in Carbohydrates

- A glycosidic bond is a covalent bond formed between a carbohydrate molecule and another molecule.

- In this reaction, the hydroxyl group (OH) of the carbohydrate combines with the hydrogen of another organic molecule, releasing a molecule of water and forming a covalent bond.

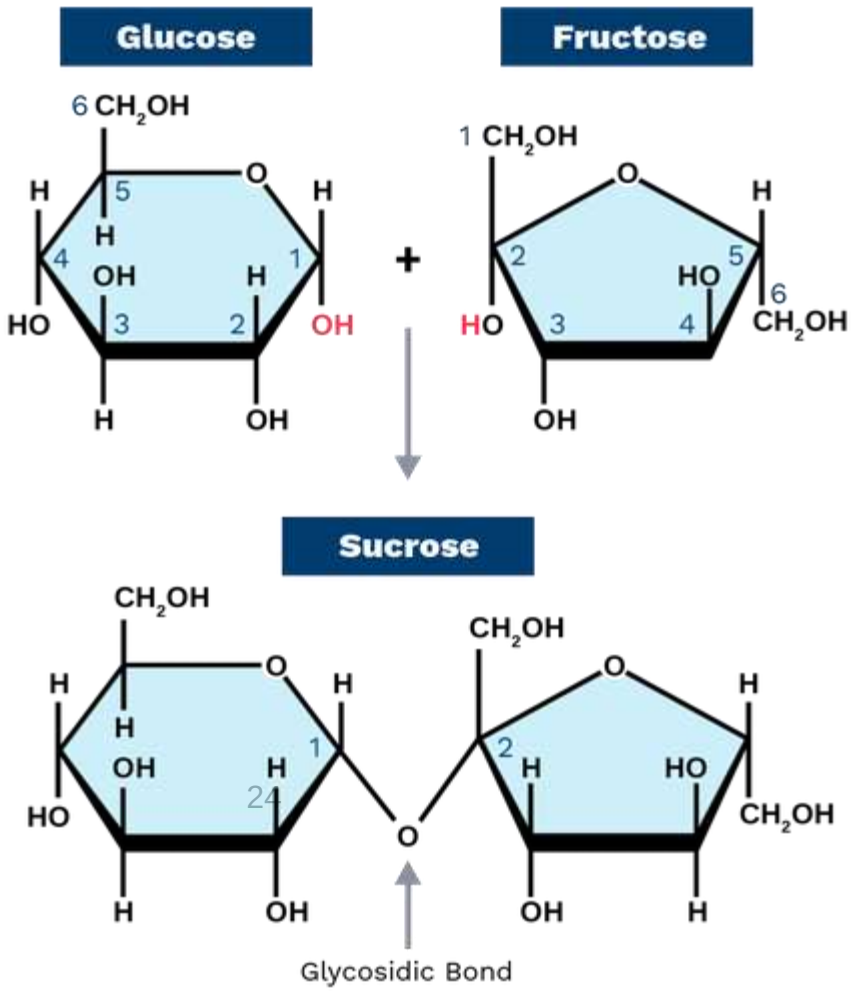


Fig: Sucrose is formed when a monomer of glucose and a monomer of fructose are joined in a dehydration reaction

Lipids

- Lipids are biomolecules that contain hydrocarbons and make up the basic building blocks of biological membranes. They have various function in living cells.
- Lipids are not soluble in water as they are non-polar, but are soluble in non-polar solvents such as chloroform.
- Examples of lipids include fats, oils, waxes, certain vitamins (such as A, D, E and K), hormones etc.
- The functions of lipids include storing energy, signalling, and acting as structural components of cell membranes.
- 1 gm of lipid (in any form) = 9 kcal.

Ester Bond in Lipids

The bond that forms between the oxygen and carbon atoms is then known as an ester bond or ester linkage.

- **Glycerol** is a small organic molecule with **three hydroxyl (OH) groups**, while a **fatty acid** consists of a **long hydrocarbon chain** attached to a **carboxyl group.**
- The basic unit of lipids are triglycerides. A triglyceride is formed when 1 glycerol molecule links with 3 fatty acid molecules by means of ester bond (covalent bond) and release molecules of H_2O

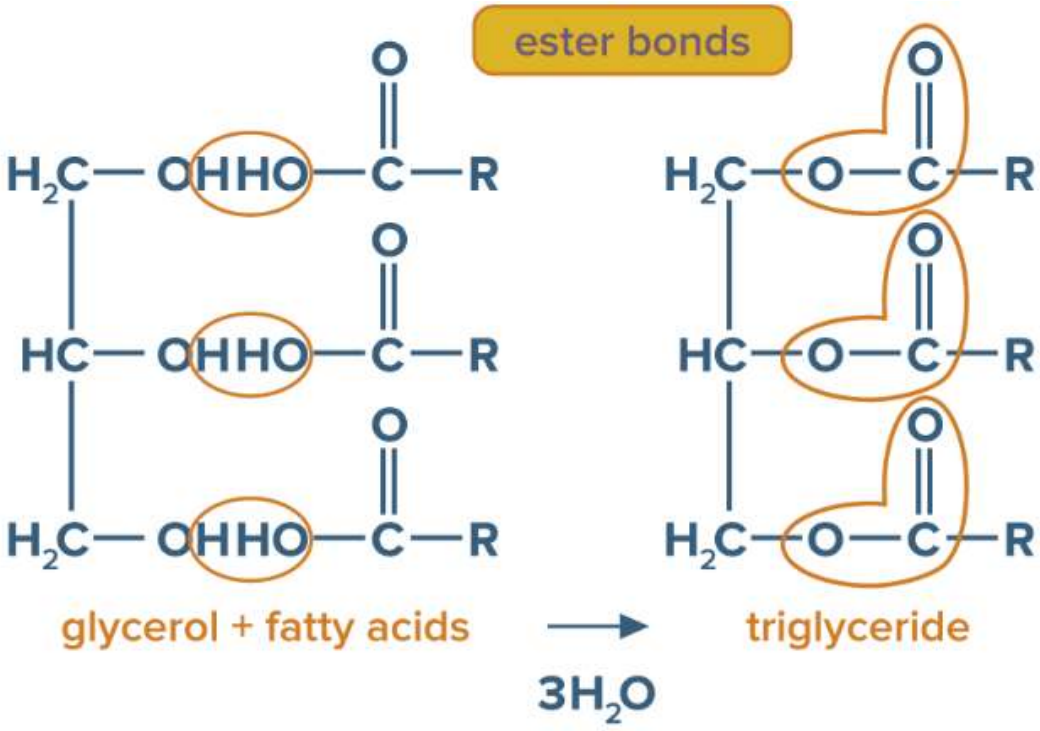
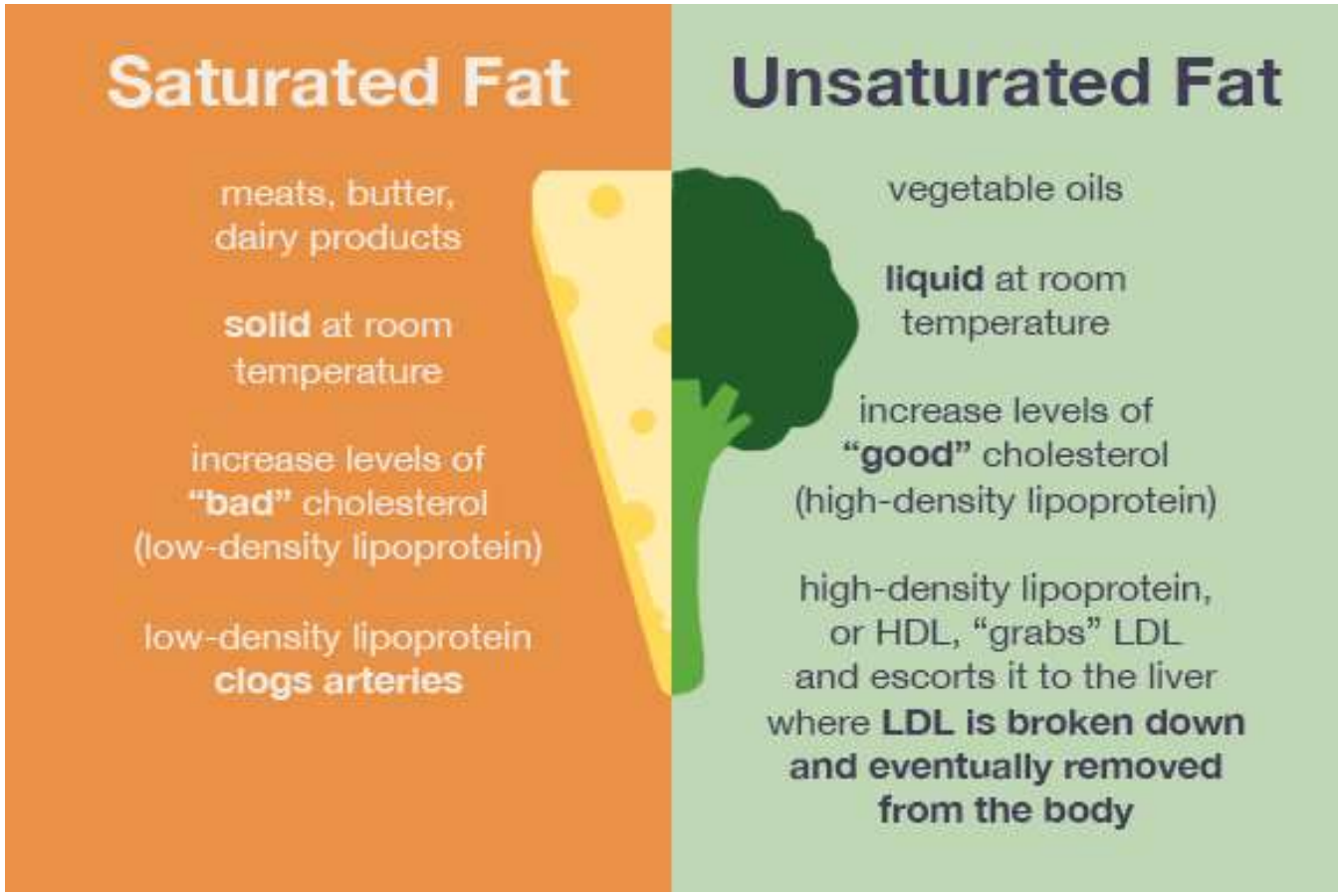


Fig: The glycerol molecule (having three OH− groups) links with three fatty acids via an ester bond

Saturated VS Unsaturated Fats



Saturated	Unsaturated
H H | | -C-C- | | H H	H H | | -C=C-
Carbon-Carbon Single Bond	Carbon-Carbon Double Bond

Proteins

- A protein is a biomolecule composed of amino acids joined together by peptide bonds.
- An amino acid is a molecule consisting of the basic amino group (NH2), the acidic carboxylic group (COOH), a hydrogen atom, and an organic side group (R) attached to the carbon atom.
- Proteins within a cell have many functions, including building cellular structures and serving as enzyme catalysts for cellular chemical reactions that give cells their specific characteristics.
- They help in metabolism by providing structural support and by acting as enzymes, carriers and hormones.
- 1 gram of Protien ≈ 4 Kcal.

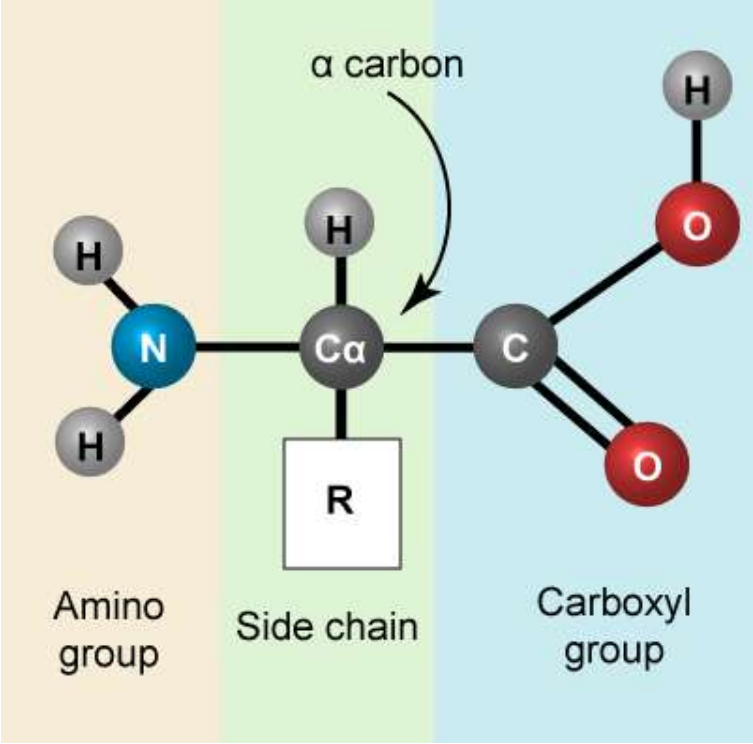


Fig- Amino Acid Structure

Peptide Bond in Protein

- Within a protein, multiple amino acids are linked together by peptide bonds, thereby forming a long chain.

- A peptide bond is a chemical bond formed between two molecules when the carboxyl group of one molecule reacts with the amino group of the other molecule, releasing a molecule of water (H2O).

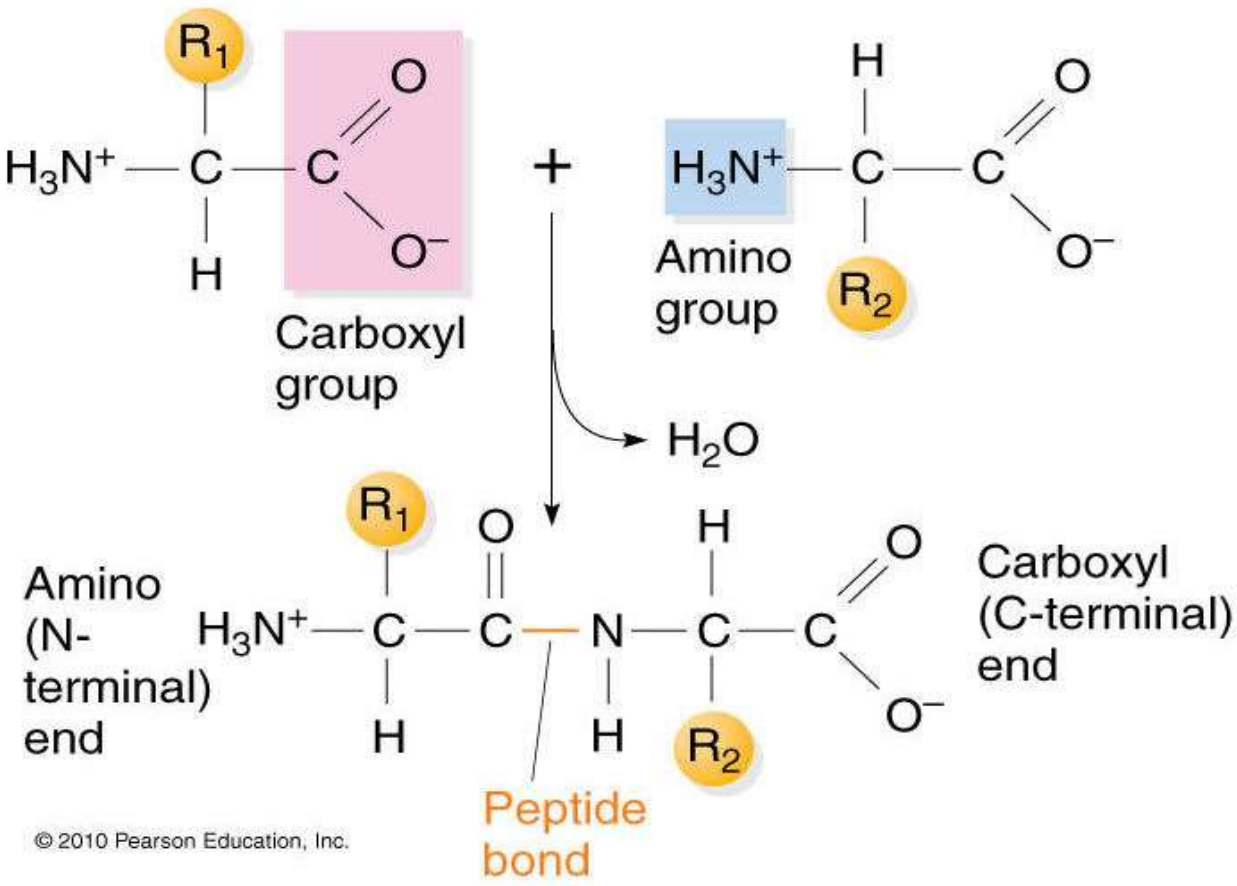
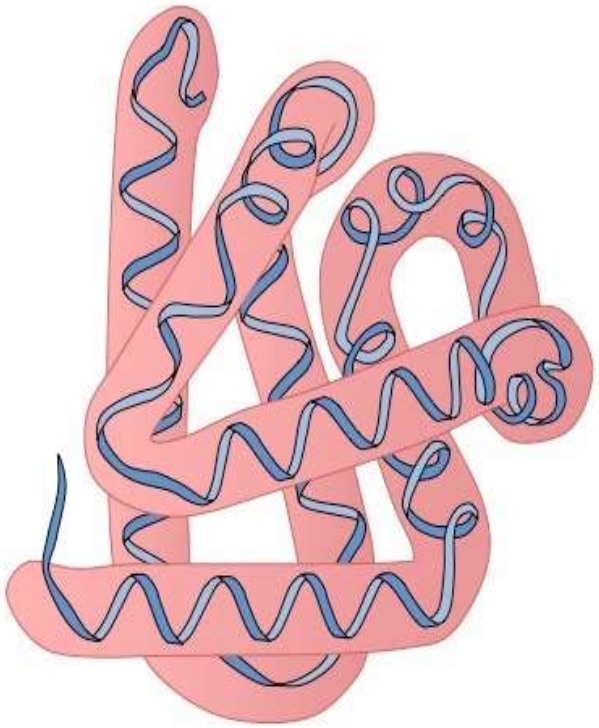


Fig- Formation of a peptide bond

Function of the Protein

	Fibrous	Globular
Shape	Long and narrow	Round / spherical
Purpose	Structural	Functional
Acid Sequence	Repetitive amino acid sequence	Irregular amino acid sequence
Durability	Less sensitive to changes in pH, temperature, etc.	More sensitive to changes in pH, temperature, etc.
Examples	Collagen, myosin, fibrin, actin, keratin, elastin	Enzymes, haemoglobin, insulin, immunoglobulin
Solubility	(Generally) insoluble in water	(Generally) soluble in water

Fibrous Protein



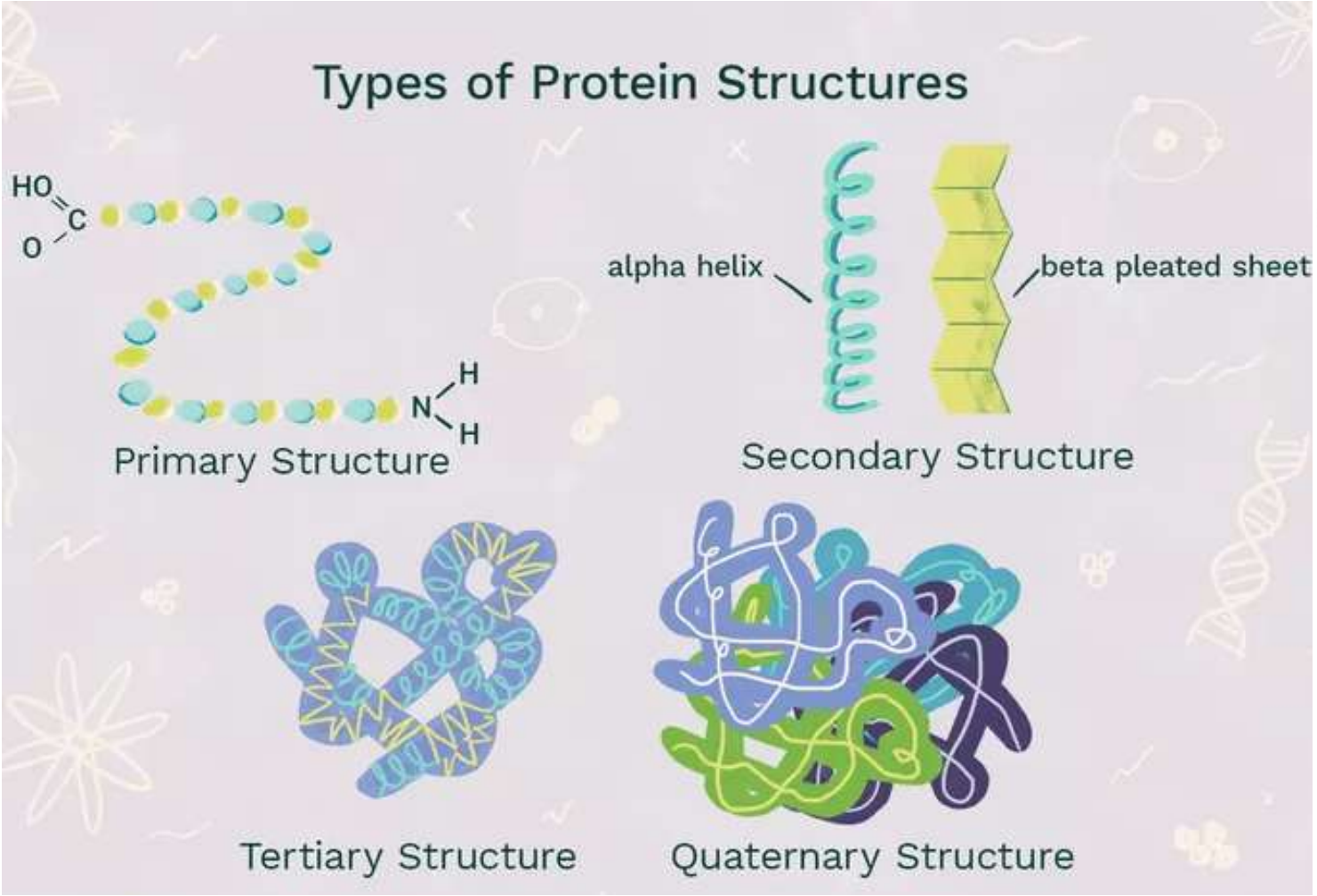
Globular Protein

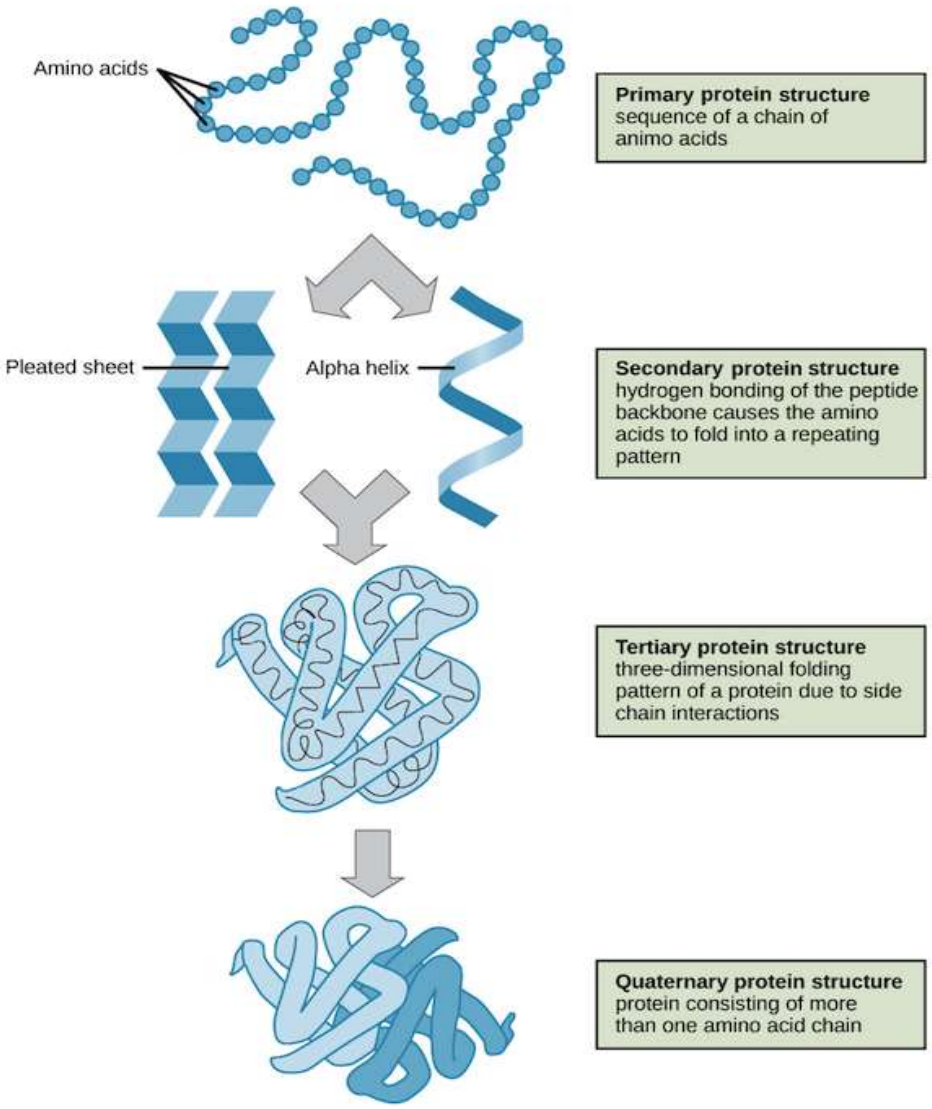
Understanding Protein Structure

- Proteins are complex molecules made of amino acids.
- Their structure determines their function in the body.

- Levels of Protein Structure:
- Primary: Amino acid sequence
- Secondary: Alpha helices and beta sheets
- Tertiary: 3D folding of the protein
- Quaternary: Multiple protein chains interacting

Structure of Protein





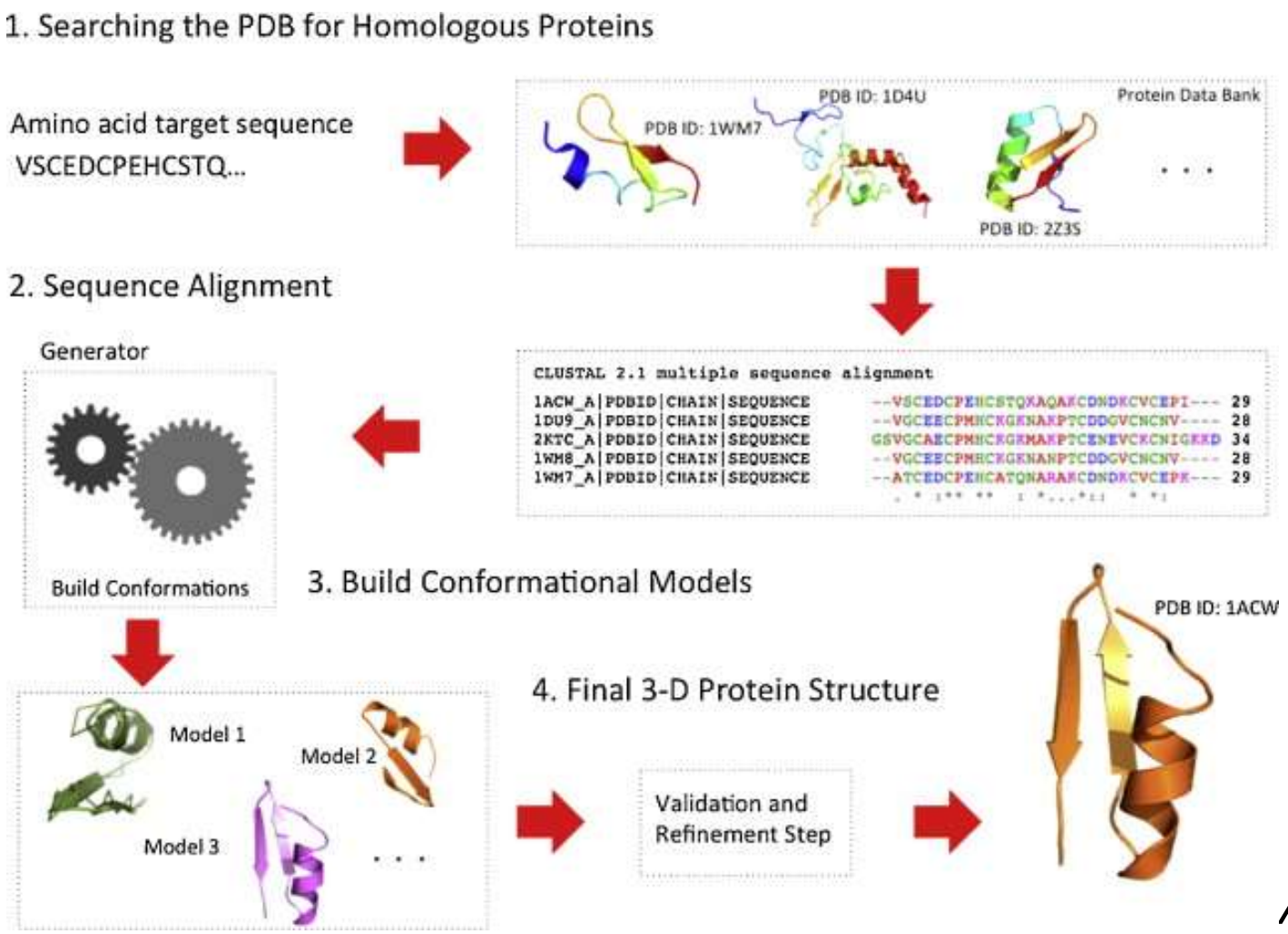
Protein Structure Prediction

Protein structure prediction is the process of determining the 3D structure of a protein from its amino acid sequence.

- Helps understand protein function
- Key to drug discovery and design
- Solves biological mysteries (e.g., enzymes, receptors)

AlphaFold, Rosetta

Protein Structure Prediction



AlphaFold, Rosetta

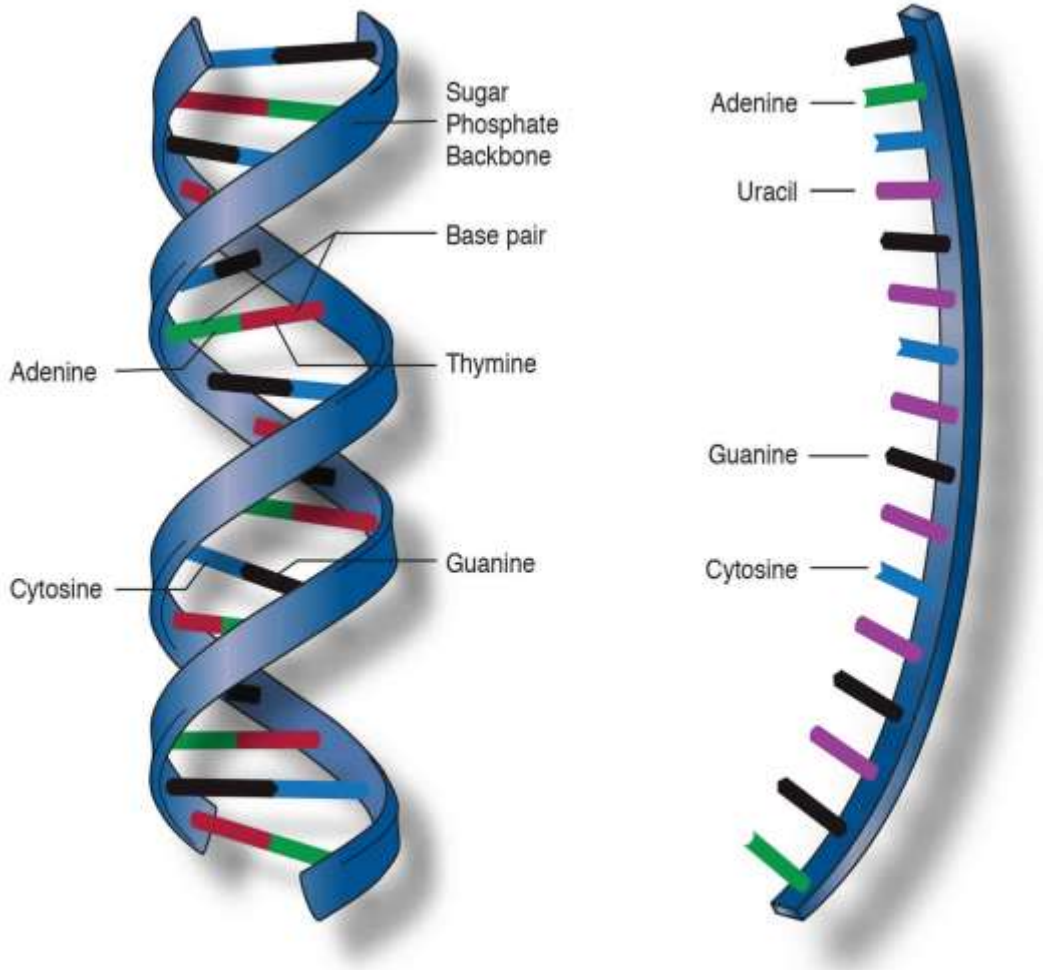
Importance of Protein Structure in Health

Correct protein folding is essential for biological function.
Misfolded proteins are linked to diseases:
- Alzheimer's (amyloid beta)
- Parkinson’s (alpha-synuclein)
- Sickle cell anemia (hemoglobin mutation)

Studying protein structures helps:
- Design treatments
- Understand disease mechanisms
- Develop vaccines (e.g., COVID-19 spike protein)

Nucleic Acids

- Nucleic acids are essential for all forms of life, and it is found in all cells.
- The functions of nucleic acids have to do with the storage and expression of genetic information.
- The building blocks of nucleic acids are nucleotides.



Deoxyribonucleic acid (DNA)

Ribonucleic acid (RNA)

Deoxyribonucleotids

- Nucleotides that compose of DNA are called deoxyribonucleotides. The three components of a deoxyribonucleotide are a five-carbon sugar called **deoxyribose, a phosphate group and a nitrogenous base**.

- DNA stores the information needed to build and control the cell.

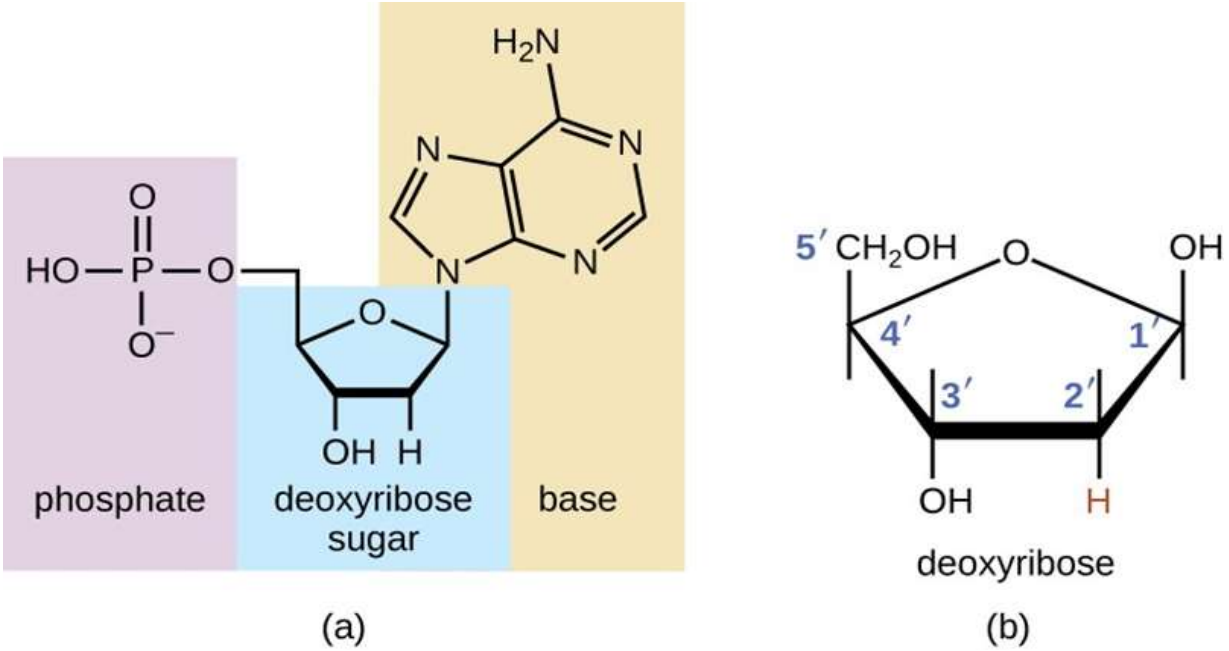


Fig: (a) deoxyribonucleotide. (b) deoxyribose sugar

Ribonucleotides

- Nucleotides that compose of RNA are called ribonucleotides. The three components of a ribonucleotide are a **pentose sugar called ribose**, a phosphate group and a nitrogenous bases (Adenine, **Uracil,** Guanine and Cytosine).

- The RNA-specific pyrimidine **uracil** forms a complementary base pair with adenine and is used **instead of the thymine** used in DNA.

- The primary role of RNA is to convert the information stored in DNA into proteins.

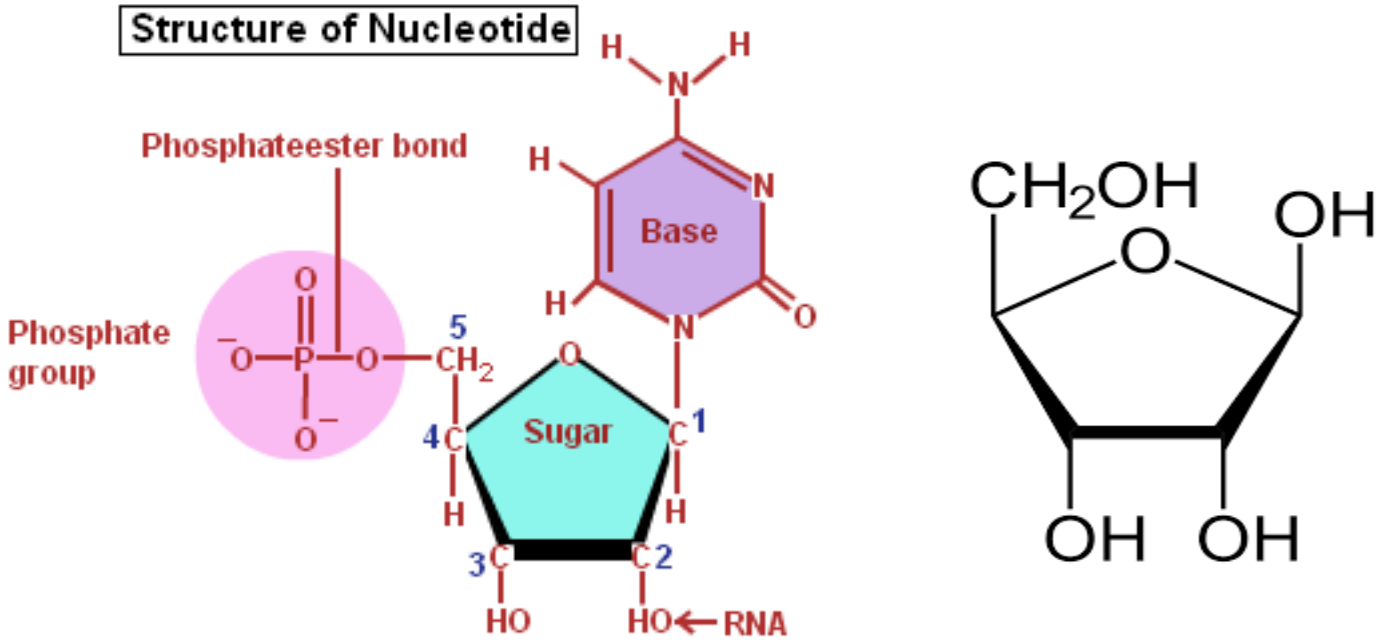


Fig: (a) ribonucleotide. (b) ribose sugar

Nitrogenous Bases

The nitrogenous bases adenine (A) and guanine (G) are the **purines**; they have a double-ring structure with a six-carbon ring fused to a five-carbon ring.

The **pyrimidines**, cytosine (C) and thymine (T), are smaller nitrogenous bases that have only a six-carbon ring structure. Thymine is unique to DNA

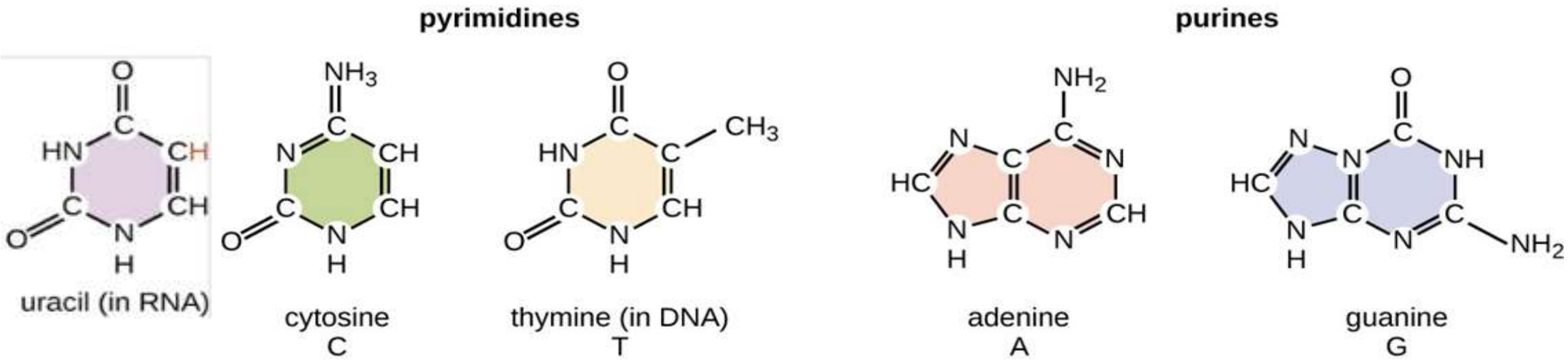


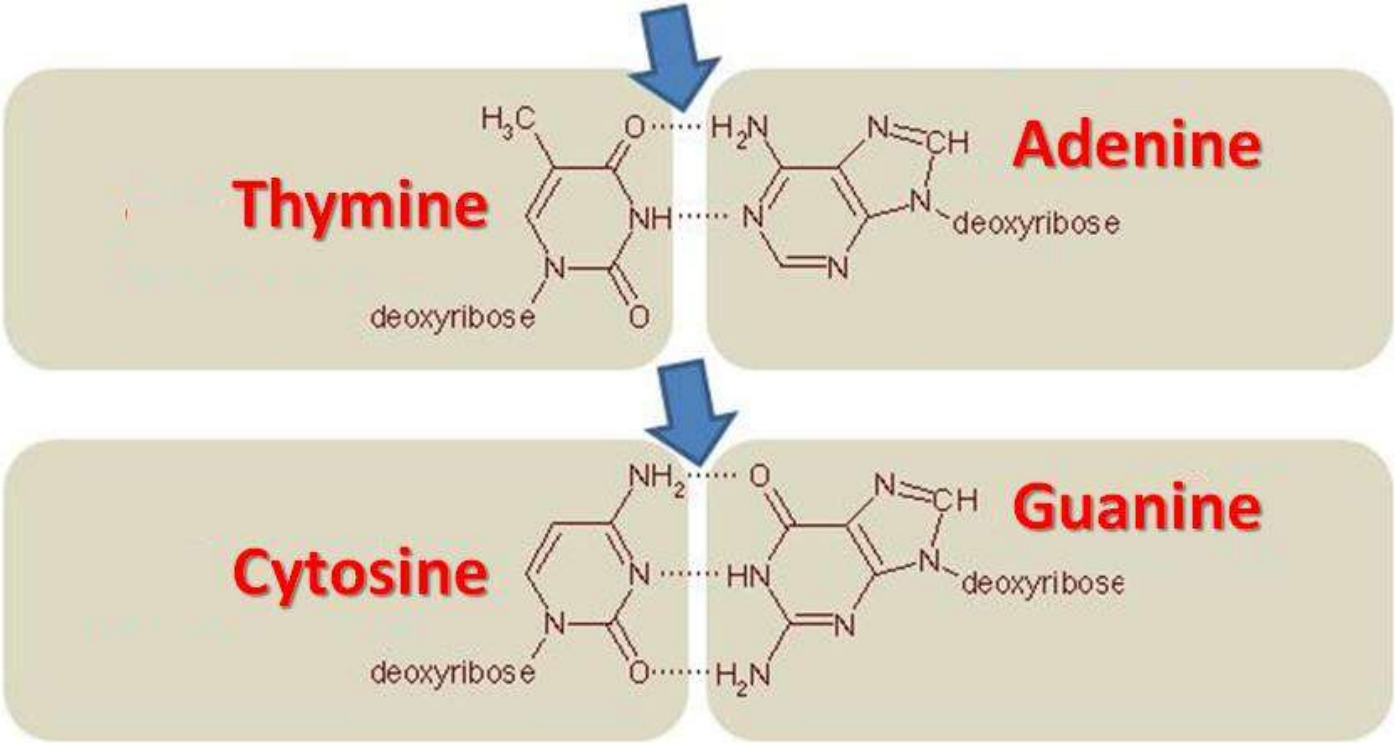
Fig- Structure of pyrimidines and purine

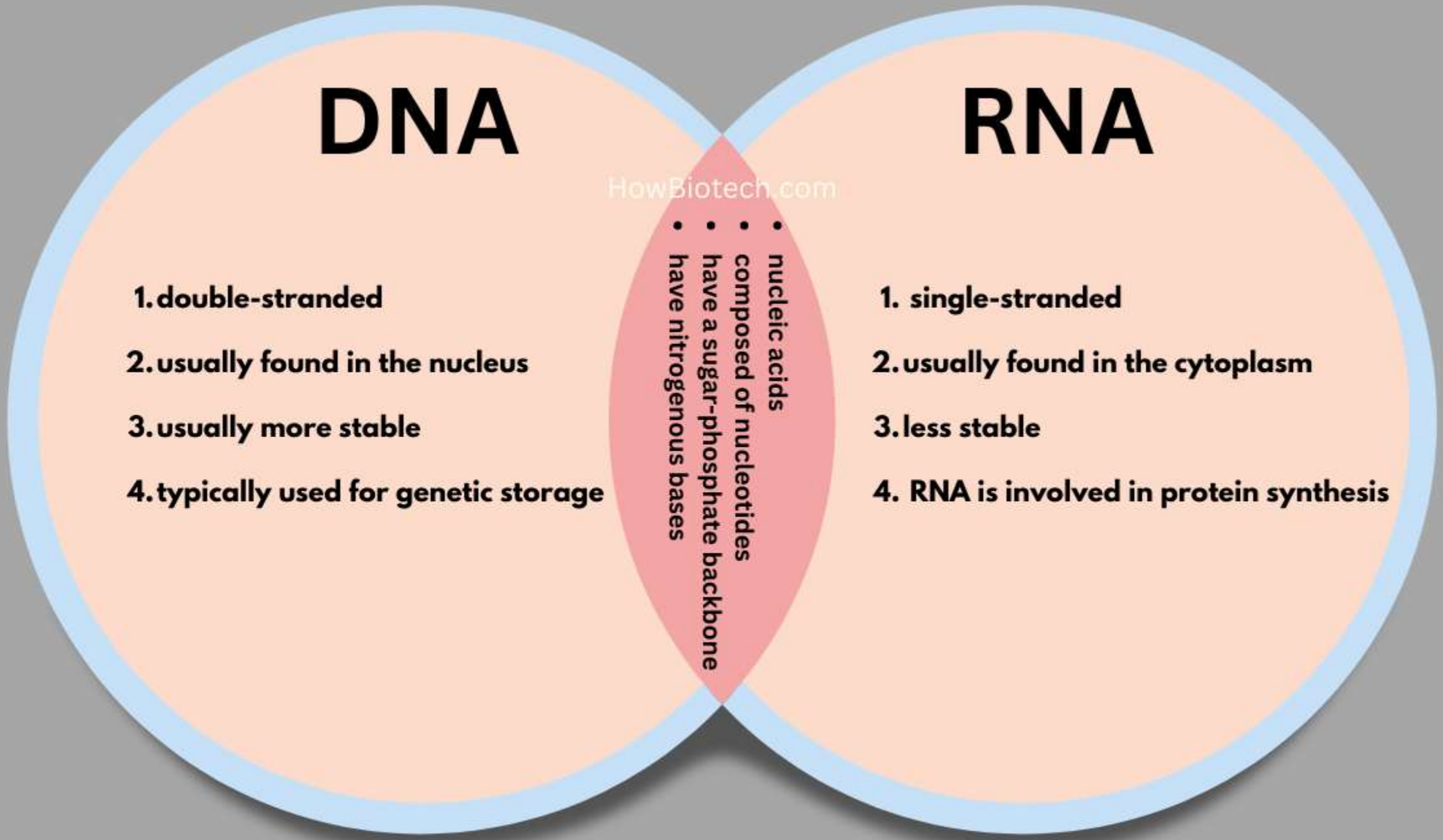
Complementary Base Pairs

The base pairs are stabilized by **hydrogen bonds**;

- In DNA, adenine (A) and thymine (T) are **complementary base pairs**, and cytosine (C) and guanine (G) are also complementary base pairs.

- Adenine and thymine form two hydrogen bonds between them,
- Cytosine and guanine form three hydrogen bonds between them.




DNA
RNA
HowBiotech.com
1. double-stranded
2. usually found in the nucleus
3. usually more stable
4. typically used for genetic storage
• nucleic acids
• composed of nucleotides
• have a sugar-phosphate backbone
• have nitrogenous bases
1. single-stranded
2. usually found in the cytoplasm
3. less stable
4. RNA is involved in protein synthesis

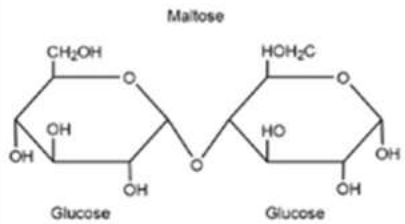
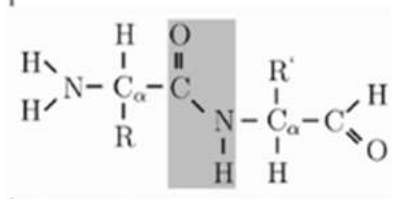
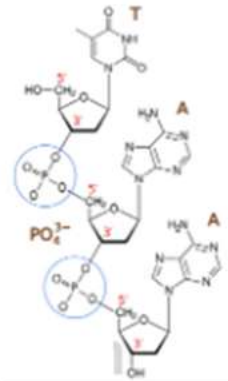
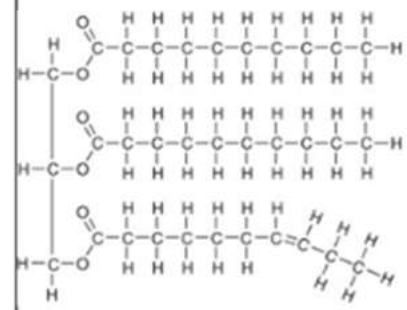
DNA vs RNA

More Information Online WWW.DIFFERENCEBETWEEN.COM

	DNA	RNA
DEFINITION	DNA is a double-stranded helix that acts as the genetic material of most living organisms.	RNA is a single-stranded nucleic acid that involves in protein synthesis.
MONOMER	Deoxyribonucleotides	Ribonucleotides
PENTOSE SUGAR	Deoxyribose	Ribose
NITROGENOUS BASES	A, T, G and C	A, U, G and C
FOLDING WITH HISTONE PROTEINS	DNA folds tightly with histone proteins and arranges into chromosomes.	Unlike DNA, RNA is not associated with histone protein.
MAIN FUNCTION	In general, DNA is the genetic material of most living organisms.	RNA helps to get the message from DNA and synthesis of proteins.
VISUALIZATION	The DNA can be analyzed by southern blotting.	RNA can be analyzed by northern blotting.
NUMBER OF STRANDS	DNA is double stranded	RNA single-stranded (single chain)
LOCATION	DNA locates inside the nucleus or in the mitochondria and chloroplasts.	RNA can be located inside the nucleus of the cell or in the cytoplasm (outside the nucleus).
PASSING TO NEXT GENERATIONS	DNA will be given to the daughter cells when a cell divides.	RNA does not inherit.

Biomolecules

COMPARISON CHART

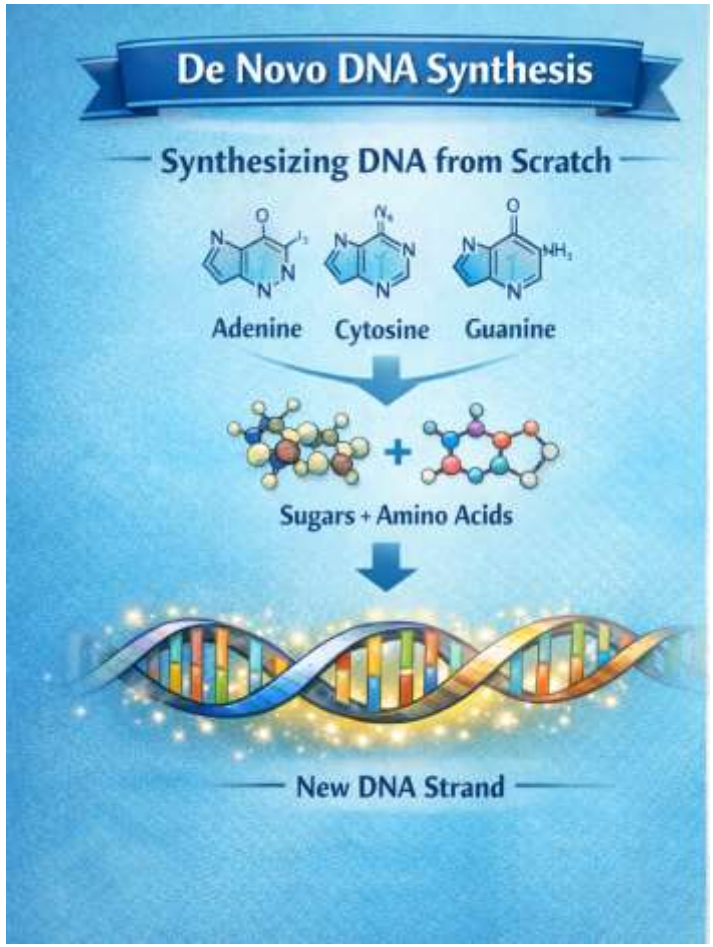
	CARBOHYDRATE	PROTEIN	NUCLEIC ACID	LIPID
Structure				
Monomers	Monosaccharides	Amino acids	Nucleotides	Fatty acids and glycerol
Primary Bond	joined by glyocosidic bond	joined by peptide bond	joined by phosphodiester bond	joined by ester bond
Elements	**C,H,O**	**C,H,O,N, S**	**C,H,O,N,P**	**C,H,O**
Function	Short term energy source Structural component Reserve food	Enzyme, structure movement, defence, hormones, Oxygen carriers	Stores genetic information	Long term energy source, insulation, biological membrane components, hormones
Examples	Starch, Glycogen, Cellulose	Insulin, Collagen, Myoglobin	DNA, RNA	Oils, Fats, Waxes

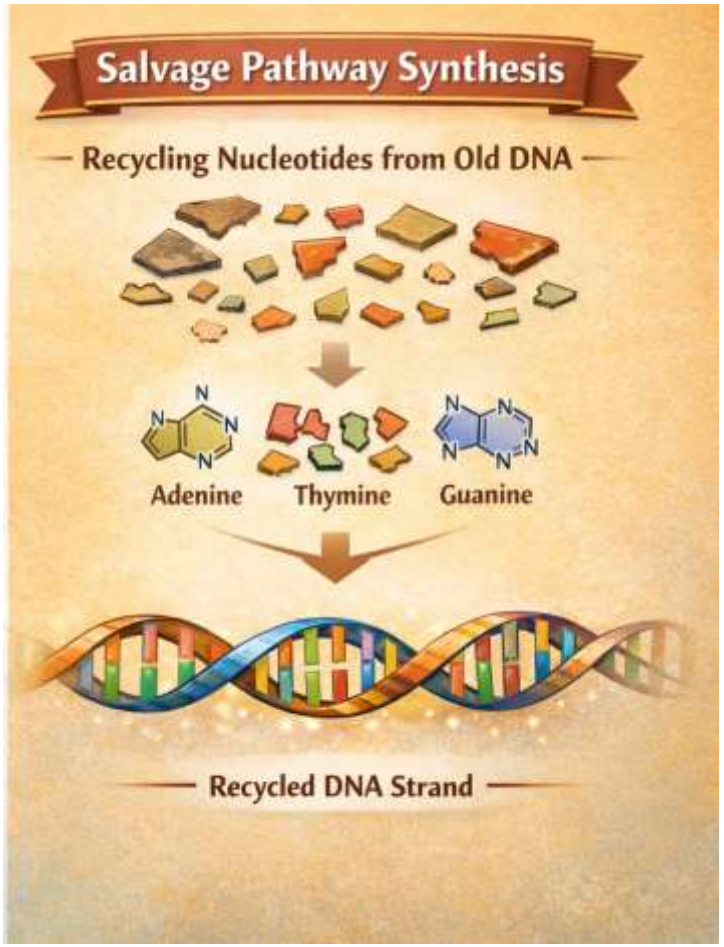
Source of Biomolecules

Macromolecule	Found In Food	Nutritional Role
Carbohydrates	Rice, bread, potatoes sugar	Quick energy, stored as glygogen
Proteins	Fish, egg, lentils (dal) meat, milk	Growth, repair, enzymes, hormones
Lipids	Ghee, oil, butter, nuts fish	Long-term energy, cell membranes, hormones
Nucleic Acids	Present in all plant/animal cells (not added in diet)	Provide geeeetic information (DNA/RNA in food gets broken down

Balanced Diet = Balanced Macromolecules

Nucleic Acids





THANK YOU